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The Spectrum of Positive Elliptic Operators and Periodic Bicharacteristics

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Introduction

Let X be a compact boundaryless C^∞ manifold and let P be a positive elliptic self-adjoint pseudodifferential operator of order $m > 0$ on X . For technical reasons we will assume that P operates on half-densities rather than functions. (We will denote the half-density bundle over X by $\Omega_{\frac{1}{2}}$.) We will also assume that P is a classical pseudodifferential operator in the sense that on every coordinate patch its total symbol $\sigma_P(x, \xi)$ admits an asymptotic expansion

$$\sigma_P(x, \xi) \sim \sum_{j=0}^{\infty} p_{m-j}(x, \xi)$$

with $p_{m-j}(x, \xi)$ homogeneous of degree $m-j$. We recall that the principal symbol p of P is equal to p_m on local coordinates, and the subprincipal symbol is equal to $p_{m-1} - \frac{1}{2i} \sum \frac{\partial^2 p}{\partial x_j \partial \xi_j}$.

Let $\lambda_1, \lambda_2, \dots$ be the eigenvalues of P . It was remarked by Chazarain in [6] and by ourselves in [11] that the sum $\sum e^{-i\sqrt{\lambda_k}t}$ is well-defined as a generalized function of t and that if T is in its singular support then the Hamiltonian vector field

$$H_q = \frac{\partial q}{\partial \xi_j} \frac{\partial}{\partial x_j} - \frac{\partial q}{\partial x_j} \frac{\partial}{\partial \xi_j}, \quad q = \sqrt[m]{p}$$

has a periodic integral curve of period T . The purpose of this article is to analyze the nature of the singularities at these T . The analysis of Hörmander [16] of the “big” singularity at $T=0$ leads to an asymptotic expansion of the form

$$\sum \rho(\mu - \mu_j) \sim (2\pi)^{-n} \sum c_k \mu^{n-1-k}, \quad \mu_j = \sqrt[m]{\lambda_j} \quad (0.1)$$

as $\mu \rightarrow +\infty$, for an appropriate class of Schwartz functions ρ . The c_k 's are the integrals over the cosphere bundle of polynomial expressions in the symbol of P and its derivatives, and are independent of ρ . (See Proposition 2.1 and (2.16).) In Section 2 we show how they are related to the residues at the poles of the zeta function of P and to the coefficients occurring in the asymptotic expansion of the trace of the heat kernel at $t=0$. From this we obtain rather easily results of Seeley [22] on the zeta function and Minakshisundaram-Pleijel [18] on the trace of the heat kernel (just for scalar operators, however). We note in passing that the asymptotic expansion of the trace of the heat kernel involves logarithmic terms unless P is a differential operator. The existence of these terms seems to have

been neglected in the literature. Section 2 concludes with a priori estimates for the spectrum which follow from (0.1) and which are used in Section 3.

There we study operators P for which all H_q solution curves are periodic with the same period. Specifically we show that if the H_q flow is periodic with period T there exists a constant β such that most of the spectrum of $\sqrt[m]{P}$ is concentrated near the lattice points $\frac{2\pi}{T}k + \beta$, $k = 1, 2, \dots$. We show that conversely if this "clustering" occurs then the H_q flow is periodic. In fact we show that if a few pathological examples are excluded then for non-periodic H_q flow the spectrum is rather equally distributed. Also the spectral estimate of Hörmander [16] can be slightly improved in this case, and an error term of order $O(\lambda^{\frac{n-1}{m}})$ replaced by an error term of order $o(\lambda^{\frac{n-1}{m}})$. See Theorem 3.5.

In Section 4 we begin our analysis of the singularities of $\sum e^{-i\sqrt[m]{\lambda_k}t}$ at periods $T \neq 0$. Our main result, Theorem 4.5, is that whenever the map $\exp TH_q: T^*X \setminus 0 \rightarrow T^*X \setminus 0$ has a clean fixed point set (in the sense of Bott), then an asymptotic expansion of the form (0.1) is valid in a neighborhood of T ; moreover, the leading term in this asymptotic expansion can be computed from such data as the length of the period and the eigenvalues of the Poincaré map. Chazarain obtains results similar to ours in [6] but without the explicit formula for the leading term. As a corollary of Theorem 4.5 we obtain the following residue formula in case all the periodic H_q solution curves of period T are isolated and non-degenerate:

$$\lim_{t \rightarrow T} (t - T) \sum e^{-i\sqrt[m]{\lambda_k}t} = \sum \frac{T_{0\gamma}}{2\pi} i^{\sigma_\gamma} |I - P_\gamma|^{-\frac{1}{2}} \quad (0.2)$$

the sum taken over all integral curves γ of period T . Here $T_{0\gamma}$ is the smallest positive period of γ , σ_γ is a Maslov factor (explained in Section 6) and P_γ the Poincaré map around γ .

For the proof of Theorem 4.5 we need some results concerning composition of Fourier integral operators under "clean intersection" assumptions, generalizing results of Hörmander [17, Chapter 4]. They are discussed in Section 5 and proved in Section 7. Similar results have been announced by Weinstein at the Conference on Fourier Integral Operators in Nice, May 1974.

If all the periodic H_q solution curves are isolated and non-degenerate and only one such curve, γ , or two such curves, γ and $-\gamma$, occur for each period¹, then from (0.2) one can determine $|I - P_\gamma^k|$ for all k . (Just apply (0.2) to the k -fold iterate of γ .) It turns out that these data almost suffice to determine P_γ itself. In fact it determines all the eigenvalues of P_γ of modulus $\neq 1$ and, up to multiplication by roots of unity, all the eigenvalues of modulus 1. This result is due to Harold Stark; and he has generously allowed us to publish it here in an appendix.

Many of the results of this paper extend to operators operating on vector bundles providing the eigenvalues of the symbol are of constant multiplicity. We hope to discuss these results in a future article. We will content ourselves here with mentioning a typical result concerning the Laplace operator on k -forms. For this

¹ This is the generic case if P is a differential operator; for pseudo-differential operators generically only one periodic solution curve occurs for each period.

operator the residue formula (0.2) is still valid except that the residue associated with γ is

$$\frac{T_{0\gamma}}{2\pi} i^{\sigma_\gamma} |I - P_\gamma|^{-\frac{1}{2}} \text{trace } H_\gamma,$$

$H_\gamma: A^k \rightarrow A^k$ being the holonomy along γ .

In conclusion we would like to thank Iz Singer and Michael Atiyah for helping us to clarify the relations among heat equation, wave equation and zeta function asymptotics; and we would like to thank Harold Stark for proving for us the result described above concerning the Poincaré map. Our main inspiration for writing this paper was the beautiful article of Hörmander [16] on the spectral function of an elliptic operator. We would also like to thank Alan Weinstein for helpful conversations concerning the material in Section 3. Formal resemblances with the methods used by Colin de Verdière [7] and Cotsaftis [8] were an incentive to the computation of some of the coefficients in the asymptotic expansions in Theorem 4.5.

1. The Operator $e^{-itP^{1/m}}$ and Its Trace

Let $Q = P^{1/m}$ be the m -th root of the unique self-adjoint extension of P in $L^2(X, \Omega_{\frac{1}{2}})$, given by the spectral theorem. According to Seeley [22], Q is a pseudo-differential operator of order 1 with principal symbol

$$q(x, \xi) = p(x, \xi)^{1/m}, \quad (1.1)$$

so again elliptic (and positive). Note that $q(x, \xi) = 1$ if and only if $p(x, \xi) = 1$ and $H_q = \frac{1}{m} H_p$ on $S^*X = \{(x, \xi) \in T^*X \setminus 0; p(x, \xi) = 1\}$. It follows that for $P = -\Delta + c$ on a Riemannian manifold X , H_q coincides on S^*X with the geodesic spray, so the projections in X of the H_q solution curves are just the geodesics with unit velocity.

As an operator acting on $\frac{1}{2}$ -densities, P has a subprincipal symbol which is an invariantly defined homogeneous C^∞ function of degree $m-1$ on $T^*X \setminus 0$ given by

$$\text{sub } P = p_{m-1} - (2i)^{-1} \sum_{j=1}^n \partial^2 p_m / \partial x_j \partial \xi_j \quad (1.2)$$

on local coordinates (Duistermaat and Hörmander [9, Section 5.2]). We have

$$\text{sub } (P^\vartheta) = \vartheta p^{\vartheta-1} \cdot \text{sub } P, \quad (1.3)$$

in particular $\text{sub } Q = 0$ if and only if $\text{sub } P = 0$. For ϑ equal to a positive integer (1.3) follows from

$$\text{sub } (A \circ B) = (\text{sub } A) \cdot b + a \cdot (\text{sub } B) + (2i)^{-1} \{a, b\}, \quad (1.4)$$

here a , resp. b are the principal symbols of A , resp. B . (1.4) is an immediate consequence of the formula

$$\sigma_{AB}(x, \xi) \sim \sum_{k=0}^{\infty} \frac{1}{k!} \left(i^{-1} \sum_j \frac{\partial^2}{\partial y_j \partial \eta_j} \right)^k \sigma_A(x, \eta) \cdot \sigma_B(y, \xi) \Big|_{\eta=\xi}^y=x$$

for the total symbol of the product of pseudo-differential operators on local coordinates. (1.3) for all positive integers ϑ implies (1.3) for all rational ϑ and then for all complex ϑ by analytic continuation.

Using the formula

$$\sigma_{A^*}(x, \xi) \sim \sum_{k=0}^{\infty} \frac{1}{k!} \left(i^{-1} \sum_j \frac{\partial^2}{\partial x_j \partial \xi_j} \right)^k \overline{\sigma_A(x, \xi)}$$

for the total symbol of the adjoint (on local coordinates) it follows that $\text{sub } A$ is real if $A = A^*$. Using the proof of [9, Proposition 5.2.1] one obtains for self-adjoint A :

$$\begin{aligned} (A(e^{i\tau\varphi} w), e^{i\tau\varphi} w) &= \tau^\alpha \cdot \int a(x, d\varphi(x)) \cdot |w(x)|^2 \\ &\quad + \tau^{\alpha-1} \int (\text{sub } A)(x, d\varphi(x)) \cdot |w(x)|^2 + O(\tau^{\alpha-2}) \quad \text{as } \tau \rightarrow \infty. \end{aligned} \quad (1.5)$$

Here α , resp. a are the order, resp. principal symbol of A . φ is a real-valued C^∞ function on X , $d\varphi(x) \neq 0$ for $x \in \text{supp } w$. Note that $|w(x)|^2$ is a density of order 1 in X if $w \in C^\infty(X, \Omega_{\frac{1}{2}})$, so can automatically be integrated over X . Formula (1.5) gives an alternative characterization of the subprincipal symbol of a self-adjoint pseudo-differential operator, from which it is obvious that it is real. Note also that $\text{sub } A = 0$ if A is a real self-adjoint differential operator of even order.

The operator

$$U(t) = e^{-itQ} \quad (1.6)$$

is unitary in $L^2(X, \Omega_{\frac{1}{2}})$, and using that it commutes with Q^{-s} it is also bounded as an operator in $H_{(s)}(X, \Omega_{\frac{1}{2}})$, uniformly in t , for all $s \in \mathbb{R}$ (Hörmander [16, Section 3]). Using repeated partial integrations with respect to t it follows that

$$\rho \mapsto \int_{-\infty}^{\infty} e^{-itQ} \rho(t) dt \quad (1.7)$$

is a continuous linear mapping: $\mathcal{S}(\mathbb{R}) \rightarrow C^\infty(X \times X, \Omega_{\frac{1}{2}})$. In particular

$$\text{Trace } U: \rho \mapsto \text{Trace} \int_{-\infty}^{\infty} e^{-itQ} \rho(t) dt \quad (1.8)$$

is a tempered distribution on $\mathbb{R}$. Here the trace of an operator with smooth kernel K is defined by

$$\text{Trace } K = \int_{x \in X} K(x, x) \quad (1.9)$$

see Atiyah and Bott [3, Section 7] for a justification of this terminology.

The positive elliptic operator Q on the compact manifold X has a discrete spectrum $\mu_j = \lambda_j^{1/m} \rightarrow \infty$ as $j \rightarrow \infty$ and corresponding orthonormal eigenfunctions $e_j \in C^\infty(X, \Omega_{\frac{1}{2}})$. (Use that Q^{-1} is a pseudo-differential operator of order -1 , so compact as an operator in $L^2(X, \Omega_{\frac{1}{2}})$ and $Q - \mu_j$ is hypo-elliptic.) The distribution kernel of $U(t)$ can be written as

$$U(t, x, y) = \sum_{j=1}^{\infty} e^{-it\mu_j} \cdot e_j(x) \cdot \overline{e_j(y)} \quad (1.10)$$

and therefore

$$\text{Trace } U(t) = \int_{x \in X} \sum_{j=1}^{\infty} e^{-it\mu_j} |e_j(x)|^2 = \sum_{j=1}^{\infty} e^{-it\mu_j} \quad (1.11)$$

is equal to the Fourier transform $\hat{\sigma}$ of the “spectral distribution”

$$\sigma(\mu) = \sum_{j=1}^{\infty} \delta(\mu - \mu_j). \quad (1.12)$$

Note that $|e_j(x)|^2$ is a density of order 1 in X and therefore can be integrated over X in an invariant way. Because Fourier transformation is an isomorphism: $\mathcal{S}'(\mathbb{R}) \rightarrow \mathcal{S}'(\mathbb{R})$ the identification of Trace U with $\hat{\sigma}$ is justified, proving at the same time that σ is a tempered distribution, that is

$$\# \{j; \mu_j \leq \mu\} \leq C|\mu|^N \quad \text{for all } \mu > 0, \quad (1.13)$$

for some constants C, N . Better estimates will follow from the more detailed description of U given below.

As a distribution in $(\mathbb{R} \times X) \times X$, U can alternatively be characterized as the kernel of the operator assigning to $u_0 \in C^\infty(X, \Omega_{\frac{1}{2}})$ the solution $u \in C^\infty(\mathbb{R} \times X, \Omega_{\frac{1}{2}})$ of the following Cauchy problem:

$$i^{-1} \partial u / \partial t + Qu = 0, \quad u(0, x) = u_0(x). \quad (1.14)$$

Theorem 1.1. U is a Fourier integral operator of class $I^{-\frac{1}{2}}(\mathbb{R} \times X, X; C)$ defined by the canonical relation

$$\begin{aligned} C = \{ & ((t, \tau), (x, \xi), (y, \eta)); (x, \xi), (y, \eta) \in T^*X \setminus 0, \\ & (t, \tau) \in T^*\mathbb{R} \setminus 0, \tau + q(x, \xi) = 0, (x, \xi) = \Phi^t(y, \eta) \}. \end{aligned} \quad (1.15)$$

Here Φ^t denotes the H_q flow in $T^*X \setminus 0$.

Proof. Using [9, Section 5.3] one can find $U_0 \in I^{-\frac{1}{2}}(\mathbb{R} \times X, X; C)$ such that $(i^{-1} \partial / \partial t + Q) \circ U_0$ has a smooth kernel and $U_0(0) = \text{identity in } C^\infty(X, \Omega_{\frac{1}{2}})$. Imitating Duhamel's principle this operator can be used to solve the inhomogeneous problem

$$i^{-1} \partial u / \partial t + Qu = f, \quad u(x, 0) = 0 \quad (1.16)$$

for small $|t|$, the solution $u \in C^\infty(\mathbb{R} \times X, \Omega_{\frac{1}{2}})$ depending continuously on $f \in C^\infty(\mathbb{R} \times X, \Omega_{\frac{1}{2}})$. It follows that by adding an operator with smooth kernel, U_0 can be changed to an operator U such that $(i^{-1} \partial / \partial t + Q) \circ U = 0$ for small $|t|$ and still $U(0) = \text{identity in } C^\infty(\mathbb{R} \times X; \Omega_{\frac{1}{2}})$. The local existence theorem for (1.16) also leads to a local uniqueness theorem for (1.14) using a standard duality argument (note that Q is self-adjoint here). In view of $U(t_1 + t_2) = U(t_1) \circ U(t_2)$ and using the local uniqueness the local solutions automatically piece together to a global solution U which differs from U_0 by an operator with a smooth kernel and therefore is of class $I^{-\frac{1}{2}}(\mathbb{R} \times X, X; C)$.

For more details, see for instance [10, Chapter 5.1]. A technical complication here is that $\square = i^{-1} \partial / \partial t + Q$ is not a pseudo-differential operator in $\mathbb{R} \times X$ because is not pseudo-local there, as observed in Nirenberg [19, Section 9]. However its wave front relation $WF'(\square)$ outside the diagonal in $T^*(\mathbb{R} \times X) \setminus 0 \times T^*(\mathbb{R} \times X) \setminus 0$ (where $\square$ is pseudo-differential) only contains points of the form $((t, x), (\tau, 0)), ((t, y), (\tau, 0))$. In taking the product with U_0 these points do not contribute to $WF'(\square \circ U_0)$ (Hörmander [17, Theorem 2.5.15]) so in the construction of U_0 (and U) we may treat Q as if it were a pseudo-differential operator in $\mathbb{R} \times X$.

Theorem 1.1 implies that $U(t): C^\infty(X, \Omega_{\frac{1}{2}}) \rightarrow C^\infty(X, \Omega_{\frac{1}{2}})$ is a Fourier integral operator of order 0 defined by the canonical transformation Φ^t (for each $t \in \mathbb{R}$). This characterization sounds more natural, but the formulation of Theorem 1.1 is slightly more precise. More explicitly, Theorem 1.1 means that $U(t, x, y)$ is a locally finite sum of integrals (converging in distribution sense) of the form

$$I(t, x, y) = (2\pi)^{-n} \int e^{i\varphi(t, x, y, \zeta)} a(t, x, y, \zeta) d\zeta. \quad (1.17)$$

Here φ is a non-degenerate phase function as in Hörmander [17, Section 1.2], $\zeta \in \mathbb{R}^n \setminus \{0\}$ and

$$d_\zeta \varphi(t, x, y, \zeta) = 0 \Rightarrow ((t, d_t \varphi), (x, d_x \varphi), (y, -d_y \varphi)) \in C. \quad (1.18)$$

The amplitude a is a symbol of order 0, having an asymptotic expansion of the form

$$a \sim \sum_{j=0}^{\infty} a_{-j}, \quad (1.19)$$

with a_{-j} homogeneous in ζ of degree $-j$. That we never need more than n frequency variables $\zeta = (\zeta_1, \dots, \zeta_n)$ follows from the fact that the tangent space of C never intersects the tangent space of the fiber in $T^*(\mathbb{R} \times X \times X)$ more than n -dimensional. (See Hörmander [17, Theorem 3.1.4 and the integration away of frequency variables in Section 3.2].)

From the construction of U it follows that the principal symbol of U is nowhere zero, implying that $WF'(U) = C$ and consequently $\text{sing supp } U = \{(t, x, y) \in \mathbb{R} \times X \times X; (x, \xi) = \Phi^t(y, \eta) \text{ for some } \xi \in (T_x X)^* \setminus \{0\}, \eta \in (T_y X)^* \setminus \{0\}\}$. Noting that Φ^t commutes with multiplication in the fibers with positive scalars because $q(x, \xi)$ is homogeneous of degree 1 in ξ , this description of $\text{sing supp } U$ does not change if we restrict $(x, \xi), (y, \eta)$ to S^*X . So for $Q = (-\Delta + c)^{\frac{1}{2}}$ on a Riemannian manifold the singular support of U is equal to the set of $(\pm t, x, y) \in \mathbb{R} \times X \times X$ such that x and y can be joined by a geodesic of length t .

Using our more detailed knowledge of U we can also give another interpretation of the distribution $\hat{\sigma} = \text{Trace } U$. If Δ denotes the diagonal map $(t, x) \mapsto (t, x, x): \mathbb{R} \times X \rightarrow \mathbb{R} \times X \times X$, then the pull-back Δ^* of functions on $\mathbb{R} \times X \times X$ to functions on $\mathbb{R} \times X$ is a Fourier integral operator (of order $\frac{1}{4}n$) defined by the canonical relation

$$WF'(\Delta^*) = \{((t, \tau), (x, \xi + \eta)), ((t, \tau), (x, \xi), (x, \eta))\}. \quad (1.20)$$

Because $\tau \neq 0$ when $((t, \tau), (x, \xi), (x, \eta)) \in WF(U)$ we can apply Hörmander [17, Theorem 2.5.11'] to find that Δ^*U is a well-defined distribution in $\mathbb{R} \times X$ and

$$WF(\Delta^*(U)) \subset \{((t, \tau), (x, \xi - \eta)); \tau + q(x, \xi) = 0, (x, \xi) = \Phi^t(x, \eta)\}. \quad (1.21)$$

Let next Π denote the projection $(t, x) \mapsto t: \mathbb{R} \times X \rightarrow \mathbb{R}$. Then integration over x is equal to the pushforward $\Pi_* = \text{transposed operator of } \Pi^*$, so it is a Fourier integral operator (of order $\frac{1}{2} - \frac{1}{4}n$) defined by the canonical relation

$$WF'(\Pi_*) = \{((t, \tau), ((t, \tau), (x, 0)))\}. \quad (1.22)$$

Applying the same theorem it follows that $\hat{\sigma} = \text{Trace } U = \Pi_*(\Delta^*(U))$ is a well-defined distribution in $\mathbb{R}$ and

$$WF(\hat{\sigma}) \subset \{(t, \tau); \tau < 0 \text{ and } (x, \xi) = \Phi^t(x, \xi) \text{ for some } (x, \xi)\}. \quad (1.23)$$

In particular this proves

Corollary 1.2. $\hat{\sigma}$ is C^∞ in the complement of the set of periods of periodic H_q solution curves in $T^*X \setminus 0$. (For $Q = (-\Delta + c)^{\frac{1}{2}}$ on a Riemannian manifold this means that the singular support of $\hat{\sigma}$ is contained in the set of periods of periodic geodesics.)

Corollary 1.2 has been proved in the same way by Chazarain [6]. A similar argument occurs in the extension of the trace to operators of the form f^*P with f a differentiable mapping transversal to the identity and P a pseudo-differential operator (acting on vector bundles) in Atiyah and Bott [3]. It might also be noted that up to this point the calculus of Fourier integral operators is not really needed because we only used that $WF'(U) = C$ and this follows already from the theorem on propagation of singularities in terms of wave front sets of Hörmander [26]. If P is a real-analytic operator on a real-analytic manifold X then the same results for U and $\hat{\sigma}$ hold with “singular support” replaced by “analytic singular support”. This follows from the theorem on propagation of singularities of Sato [20], Sato, Kawai and Kashiwara [21, Section 2.1]. (A proof of this theorem not using the sheaf $\mathcal{C}$ has been given by Andersson [1]).

A more explicit proof of Corollary 1.2 can be given using the local representations (1.17) of U . It follows that for $\hat{\rho} \in C_c^\infty(\mathbb{R})$ the integral

$$(2\pi)^{-1} \int e^{i\mu t} \hat{\rho}(t) \hat{\sigma}(t) dt = \sum_{j=1}^{\infty} \rho(\mu - \mu_j) \quad (1.24)$$

is a finite sum of integrals of the form

$$\begin{aligned} (2\pi)^{-1-n} \iiint e^{i\mu t} e^{i\varphi(t, x, x, \zeta)} \hat{\rho}(t) a(t, x, x, \zeta) d\zeta dx dt \\ = (2\pi)^{-1-n} \mu^n \iiint e^{i\mu(t + \varphi(t, x, x, \zeta))} \hat{\rho}(t) a(t, x, x, \mu\zeta) d\zeta dx dt. \end{aligned} \quad (1.25)$$

Applying the principle of stationary phase it follows that this is rapidly decreasing as $\mu \rightarrow \infty$ unless

$$\begin{aligned} d_\zeta \varphi(t, x, x, \zeta) &= 0, \\ d_x \varphi(t, x, x, \zeta) + d_y \varphi(t, x, x, \zeta) &= 0, \\ 1 + d_t \varphi(t, x, x, \zeta) &= 0 \end{aligned} \quad (1.26)$$

for some $t \in \text{supp } \hat{\rho}$, $x \in X$, $\zeta \in \mathbb{R}^n \setminus \{0\}$. In view of (1.18), (1.15) this just means that $p(x, \zeta) = 1$, $(x, \zeta) = \Phi'(x, \zeta)$ if we write $\xi = d_x \varphi(t, x, x, \zeta)$.

A priori different H_q solution curves with the same period T might lead to a cancellation of the singularity of $\hat{\sigma}$ at T . Only a more detailed examination of the asymptotic expansion of (1.24) as $\mu \rightarrow \infty$ can show, using additional geometric assumptions on the H_q flow if necessary, that such cancellation actually does not take place. See Sections 2, 4 below.

No statement like Corollary 1.2 holds if we replace the first order operator Q by a higher order operator P . For instance the function

$$f(t) = \sum_{n \in \mathbb{Z}} e^{-itn^2}$$

can be regarded as the trace of e^{-itP} for a second order operator P on the circle. On the other hand f is an automorphic function (cf. Serre [23, Chapter 7, § 1, 2

and 6]). The orbit of the origin under the fractional linear transformations

$$z \mapsto \frac{az+b}{cz+d}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z})$$

is dense on the real axis, and this shows that the singular support of f is equal to the whole real axis. This example was brought to our attention by Professor Atiyah.

2. The Singularity of $U(t, x, x)$ at $t=0$

Because every H_q solution curve is periodic with period 0, a big singularity of $U(t, x, x)$ can be expected at $t=0$. We start with a Fourier analysis of this singularity. Note that the canonical density in T^*X divided by the projection: $T^*X \rightarrow X$ defines a density in the fiber $(T_x^*X)^*$, the value of which is a density in T_x^*X . Dividing again by dq leads to a T_x^*X -density-valued density on $S_x^*X = \{\xi \in (T_x^*X)^*; q(x, \xi)=1\}$.

Proposition 2.1. *There exists a sequence $\omega_1, \omega_2, \dots$ of real valued smooth densities in X such that for every $\rho \in \mathcal{S}(\mathbb{R})$ with $\text{supp } \hat{\rho}$ contained in a sufficiently small neighborhood of 0 and $\hat{\rho}=1$ in another neighborhood of 0:*

$$\sum_j \rho(\mu - \mu_j) |e_j|^2 \sim (2\pi)^{-n} \sum_{k=0}^{\infty} \omega_k \mu^{n-k-1} \quad (2.1)$$

for $\mu \rightarrow \infty$ and rapidly decreasing as $\mu \rightarrow -\infty$, asymptotics in the $C^\infty(X, \Omega_1)$ -topology. Here

$$\omega_0(x) = \text{vol}(S_x^*X), \quad \omega_n(x) = 0, \quad (2.2)$$

and

$$\omega_1(x) = (1-n) \cdot \int_{S_x^*X} \text{sub } Q.$$

Moreover, if $P = Q^m$ is a differential operator, then

$$\omega_{n+lm}(x) = 0 \quad \text{for } l=1, 2, \dots \quad \text{and} \quad \omega_k(x) = 0 \quad \text{for } k \text{ odd.} \quad (2.4)$$

Proof. For small $|t|$, $U(t, x, y)$ can locally be represented by an integral of the form (1.17) with the phase function

$$\varphi(t, x, y, \eta) = \psi(x, y, \eta) - t \cdot q(y, \eta) \quad (2.5)$$

introduced by Hörmander [16, Section 3]. Here $\psi(x, y, \eta)$ is, as a function of the first variable, the local solution near $x=y$ of the Cauchy problem

$$\begin{aligned} q(x, d_x \psi(x, y, \eta)) &= q(y, \eta), \\ \psi(x, y, \eta) &= 0 \quad \text{when} \quad \langle x - y, \eta \rangle = 0, \end{aligned} \quad (2.6)$$

and finally

$$d_x \psi(x, y, \eta) = \eta \quad \text{for } x = y.$$

Substituting $\eta = \mu \cdot \omega \cdot \tilde{\eta}$, $\omega > 0$, $q(y, \tilde{\eta}) = 1$, and treating the integration over t , ω with the method of stationary phase,

$$(2\pi)^{-1} \int e^{i\mu t} \hat{\rho}(t) U(t, y, y) dt = (2\pi)^{-n} \mu^{n-1} \int_{S_y^*X} b(y, \tilde{\eta}, \mu) d\tilde{\eta},$$

with

$$b(y, \tilde{\eta}, \mu) \sim \sum_{r=0}^{\infty} \frac{1}{r!} \mu^{-r} \left(i^{-1} \frac{\partial^2}{\partial t \partial \omega} \right)^r [\hat{\rho}(t) a(t, y, y, \mu \omega \tilde{\eta}) \cdot \omega^{n-1}]_{\substack{t=0 \\ \omega=1}}$$

as $\mu \rightarrow \infty$, uniformly in $(y, \tilde{\eta}) \in S^*X$. Here the integral over the unbounded t, ω -domain is split into one over a compact neighborhood of $t=0, \omega=1$ and a domain without stationary points for the phase function, using a partition of unity. The latter integral is shown to be rapidly decreasing using repeated partial integrations as in Hörmander [17, Section 1.2]. This proves (2.1) with

$$\begin{aligned} \omega_k(y) = & \int_{S_x^*X} a_{-k}(0, y, y, \tilde{\eta}) d\tilde{\eta} + \sum_{r=1}^k (n-k+r-1) \dots (n-k) \\ & \cdot \int_{S_x^*X} \frac{1}{r!} (i^{-1} \partial/\partial t)^r a_{-k+r}(0, y, y, \tilde{\eta}) d\tilde{\eta}. \end{aligned} \quad (2.7)$$

If μ is replaced by $-\mu$ then the phase function has no stationary points for $\omega > 0$ and we obtain rapid decrease in (2.1) as $\mu \rightarrow -\infty$, reflecting the positivity of the spectrum.

In order to compute the ω_k , observe first that we need

$$(2\pi)^{-n} \int e^{i\psi(x, y, \eta)} a(0, x, y, \eta) d\eta = U(0, x, y) = \delta(x - y). \quad (2.8)$$

From (2.6) it follows that $\psi(x, y, \eta)$ is a non-degenerate phase function locally defining the same Lagrange manifold as $\langle x - y, \eta \rangle$, and $d_\eta^2 \psi = d_\eta^2 \langle x - y, \eta \rangle = 0$ at $x = y$. So according to Hörmander [17, Theorem 3.1.6] there exists locally a homogeneous change of coordinates $\eta = \eta(x, y, \zeta)$ such that $\eta(y, y, \zeta) = \zeta$ and

$$\psi(x, y, \eta(x, y, \zeta)) = \langle x - y, \zeta \rangle. \quad (2.9)$$

Substituting this in the left hand side of (2.8) we see that $U(0) = I$ if we choose

$$a(0, x, y, \eta(x, y, \zeta)) = 1/|\det d_\zeta \eta(x, y, \zeta)|. \quad (2.10)$$

In particular $a_0(0, y, y, \eta) = 1$ and $a_{-j}(0, x, y, \eta) = 0$ for all $j > 0$, implying (2.2).

Applying $i^{-1} \partial/\partial t + Q$ to (1.17) under the integral sign amounts to replacing a by an amplitude a' such that

$$\begin{aligned} a'(t, x, y, \eta) \sim & \frac{1}{i} \frac{\partial a}{\partial t}(t, x, y, \eta) - q(y, \eta) \cdot a(t, x, y, \eta) \\ & + \sum_k \frac{1}{k!} \left(i^{-1} \sum_j \frac{\partial^2}{\partial \tilde{x}_j \partial \tilde{\xi}_j} \right)^k [Q(x, \chi(\tilde{x}, x, y, \eta) + \tilde{\xi}) a(t, x + \tilde{x}, y, \eta)]_{\substack{\tilde{x}=0 \\ \tilde{\xi}=0}} \end{aligned}$$

for $|\eta| \rightarrow \infty$, here we have written

$$\psi(x + \tilde{x}, y, \eta) = \psi(x, y, \eta) + \langle \tilde{x}, \chi(\tilde{x}, x, y, \eta) \rangle. \quad (2.11)$$

The equation

$$a'_{-j}(t, x, y, \eta) = 0 \quad (2.12)$$

is a first order linear partial differential equation for a_{-j} involving only the a_{-k} for $k < j$. With the initial condition (2.10) these successive “transport equations”

for the a_{-j} have a locally unique solution, leading to a local description of the solution $\bar{U}$ of $(i^{-1} \partial/\partial t + Q) U = 0$, $U(0) = I$, modulo integral operators with smooth kernels. Differentiating (2.12) sufficiently many times with respect to t at $t=0$, $x=y$, and using (2.6), (2.9), (2.10), (2.11), the $\omega_k(y)$ in (2.7) can in principle be computed explicitly, although the computations become rapidly complicated for increasing k . For $j=0$ this procedure leads to $i^{-1} \partial a_0/\partial t(0, y, y, \eta) = -(\text{sub } Q)(y, \eta)$, proving (2.3).

A function f on S^*X will be called even if $f(x, -\xi) = f(x, \xi)$, and odd if $f(x, -\xi) = -f(x, \xi)$. If $P = Q^m$ is a differential operator then m is even, and the hypothesis that the homogeneous part of degree j in $\sigma_Q(x, \xi)$ is an even, resp. odd function on S^*X if j is odd resp. even leads to the correct conclusion that the homogeneous part of degree k in $\sigma_P(x, \xi)$ is even, resp. odd if k is even, resp. odd. Because $\sigma_Q(x, \xi)$ is determined locally in S^*X by $\sigma_P(x, \xi)$, the hypothesis gives the correct solution for $\sigma_Q(x, \xi)$ at the antipodal points. By induction with respect to r it follows from (2.10), (2.12) that $(\partial/\partial t)^r a_{-j}(0, y, y, \eta)$ is an even, resp. odd function on S^*X if $r-j$ is even, resp. odd, so (2.7) implies that $\omega_k = 0$ if k is odd.

Finally the assumption that P is a differential operator has the effect that applying it to (1.17) amounts to changing a into an amplitude a'' satisfying again that $a''_{-j}(0, x, y, \eta) = 0$ for $j > 0$. As in the proof of $\omega_n = 0$ it follows that the coefficient of μ^{-1} in the asymptotic expansion of $(2\pi)^{-1} \int e^{i\mu t} \hat{\rho}(t) P U(t, y, y) dt$ for $\mu \rightarrow \infty$ vanishes. On the other hand $P \circ U(t) = Q^m e^{-itQ} = (i\partial/\partial t)^m e^{-itQ}$, which on the inverse Fourier transform side amounts to multiplication by μ^m . So the coefficient of μ^{-1-m} in (2.1) is equal to zero, that is $\omega_{n+m} = 0$. Replacing P by P^l in this argument proves that $\omega_{n+lm} = 0$ for all positive integers l .

For $P = Q^m$, $\lambda_j = \mu_j^m$, the ζ -function

$$Z_P(s) = \sum_j \lambda_j^{-s} \cdot |e_j|^2, \quad (2.13)$$

which is the restriction of the distribution kernel of P^{-s} to the diagonal, is a holomorphic function of s for $\text{Re } s$ sufficiently large, with values in the space of smooth densities on X . (The distribution kernel of P^{-s} is continuous for $\text{Re } s$ sufficiently large and $\frac{d}{dx_i} P^{-s}(x, x)$ is equal to the restriction to the diagonal of the distribution kernel of $(\partial/\partial x_i) P^{-s} - P^{-s}(\partial/\partial x_i)$, which is of the same order $-\text{Re } s$, so there is no loss of differentiability.)

Corollary 2.2. Z_P has a meromorphic extension to the complex plane, having only simple poles at $s = (n-k)/m$ with residues equal to $(2\pi)^{-n} \omega_k/m$, $k=0, 1, 2, \dots$. Moreover Z_P is of at most polynomial growth on each half space $\text{Re } s \geq c$ excluding neighborhoods of the poles.

Proof. Let $0 < \varepsilon < \mu_1 = \text{first eigenvalue of } Q$, choose $\chi \in C^\infty(\mathbb{R})$, $\chi(\mu) = 0$ for $\mu \leq \varepsilon$, $\chi(\mu) = 1$ for $\mu \geq \mu_1$. Writing $\chi_s(\mu) = \chi(\mu) \mu^{-s}$, $\Sigma(\mu) = \sum_j \delta(\mu - \mu_j) |e_j|^2$, we have

$$Z_Q(s) = \langle \Sigma, \chi_s \rangle = \langle \hat{\Sigma}, \check{\chi}_s \rangle.$$

For $t \neq 0$ repeated partial integrations give

$$\check{\chi}_s(t) = (2\pi)^{-1} \int e^{i\mu t} \chi(\mu) \mu^{-s} d\mu = (2\pi)^{-1} (i/t)^k \int e^{i\mu t} (\partial/\partial \mu)^k [\chi(\mu) \mu^{-s}] d\mu,$$

showing that

$$|t^k \check{\chi}_s(t)| \leq C(1+|s|)^k \cdot e^{-(\operatorname{Re} s + k - 1)t} / (\operatorname{Re} s + k - 1)$$

for $\operatorname{Re} s > -k + 1$. Because this holds for all k and similar estimates hold for the derivatives of $\check{\chi}_s(t)$ with respect to t it follows that $(1 - \hat{\rho}) \check{\chi}_s$ is an entire function of s with values in $\mathcal{S}(\mathbb{R})$ having at most polynomial growth in half spaces $\operatorname{Re} s \geq c$. Here ρ is as in Proposition 2.1. As a consequence

$$\langle \Sigma, \chi_s \rangle - \langle \rho * \Sigma, \chi_s \rangle = \langle (1 - \hat{\rho}) \hat{\Sigma}, \check{\chi}_s \rangle$$

is also entire in s , and of polynomial growth in half spaces $\operatorname{Re} s \geq c$.

If $f(\mu) = O(\mu^{n-1-k})$ for $\mu \rightarrow \infty$ then $\int f(\mu) \chi(\mu) \mu^{-s} d\mu$ is holomorphic and bounded in half spaces $\operatorname{Re} s \geq c$ with $c > n - k$. On the other hand $\int \mu^{n-1-k} \chi(\mu) \mu^{-s} d\mu = \psi(s)/(s - (n - k))$, where $\psi(s) = \int \mu^{n-k-s} \frac{d\chi}{d\mu}(\mu) d\mu$ is entire in s , and bounded in half spaces $\operatorname{Re} s \geq c$, because $d\chi/d\mu$ has compact support. Moreover

$$\psi(n - k) = \int \frac{d\chi}{d\mu}(\mu) d\mu = 1.$$

So (2.1) implies that $\langle \rho * \Sigma, \chi_s \rangle$ can be extended to a meromorphic function of s having only simple poles at $s = n - k$ with residue equal to $(2\pi)^{-n} \omega_k$, and bounded in half spaces $\operatorname{Re} s \geq c$. This proves the statements for Z_Q , for Z_P they follow from $Z_P(s) = Z_Q(ms)$.

Note that Z_P has no pole at $s = 0$, and not either at $s = -1, -2, \dots$ if P is a differential operator. This proves part of the results of Seeley [22], who treats general elliptic operators on vector bundles and also gives a formula for $Z_P(0)$. The latter has been used by Atiyah, Bott and Patodi [4] for an alternative proof of the Atiyah-Singer index theorem. However they work rather with the restriction

$$\Theta_P(z) = \sum_j e^{-\lambda_j z} |e_j|^2 \quad (2.14)$$

of the distribution kernel of the “heat operator” e^{-zP} to the diagonal. This is a smooth density on X depending holomorphically on z for $\operatorname{Re} z > 0$. For $\Theta_P(z)$ we obtain:

Corollary 2.2'. *There exist smooth densities v_l , $l = 0, 1, 2, \dots$ on X such that*

$$\begin{aligned} m(2\pi)^n \Theta_P(z) \sim & \sum_{\substack{k \neq n+1m \\ l \in \mathbb{N}}} \Gamma\left(\frac{n-k}{m}\right) \cdot \omega_k \cdot z^{(k-n)/m} \\ & + \sum_{\substack{l=1 \\ n+1m \in \mathbb{Z}}}^{\infty} \frac{(-1)^{l+1}}{l!} \cdot \omega_{n+1m} \cdot z^l \cdot \log z + \sum_{l=0}^{\infty} v_l \cdot z^l \quad \text{for } z \searrow 0. \end{aligned} \quad (2.15)$$

Proof. Θ_P and Z_P are related to each other by the inverse Mellin transform

$$a) \quad \Theta_P(z) = \frac{1}{2\pi i} \int_{\operatorname{Re} s = c} z^{-s} Z_P(s) \Gamma(s) ds,$$

which holds for c sufficiently large. This follows from summing the formula

$$\begin{aligned} b) \quad e^{-\lambda z} &= \frac{1}{2\pi i} \int_{\text{Re } s=c} z^{-s} \lambda^{-s} \Gamma(s) ds \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\sigma \log(z\lambda)} (z\lambda)^{-c} \Gamma(c+i\sigma) d\sigma. \end{aligned}$$

In turn b) follows from the Fourier inversion formula if we view

$$c) \quad \Gamma(c+i\sigma) = \int_0^{\infty} t^{c+i\sigma-1} e^{-t} dt = \int_{-\infty}^{\infty} e^{i\sigma t} e^{ct-e^t} dt$$

as the Fourier transform at $-\sigma$ of the function $t \mapsto e^{ct-e^t}$.

Shifting the path of integration in c) to the line $\text{Im } \tau = \varepsilon$ one obtains the estimate

$$d) \quad |\Gamma(c+i\sigma)| \leq e^{-\sigma\varepsilon} \int e^{ct-e^t \cos \varepsilon} dt$$

for $c > 0$, $|\varepsilon| < \pi/2$. In combination with the well-known equation

$$e) \quad \Gamma(s) = \frac{1}{s(s+1)\dots(s+k-1)} \Gamma(s+k)$$

it follows that $\Gamma(s)$ is exponentially decreasing in all vertical strips, excluding neighborhoods of the poles. Combining this with the growth properties of $Z_p(s)$ it follows that $Z_p(s) \cdot \Gamma(s)$ is exponentially decreasing in vertical strips, excluding neighborhoods of the poles.

This allows to shift the path of integration in a) by letting $c \searrow -\infty$. Obviously the integral in a) is $O(z^{-N})$ for $z \searrow 0$ so in the asymptotic expansion of $\Theta_p(z)$ for $z \searrow 0$ modulo terms $O(z^n)$ for all N we only pick up the contributions from the poles which are encountered in shifting the path of integration. This leads to (2.15), the logarithmic terms occurring whenever poles of $Z_p(s)$ and $\Gamma(s)$ coincide to produce a double pole. The v_l are bijectively related to the values of Z_p at $-l$ (after taking off the pole of Z_p if necessary), apparently no further information about these quantities can be obtained from the asymptotic expansion (2.1).

(2.15) generalizes the well-known expansion of Minakshisundaram-Pleijel [18] for the Laplace operator. For $m=1$, (2.15) can also be proved by viewing $\sum e^{-i\mu_j t} |e_j|^2$ as the boundary value (in distribution sense) for $z \rightarrow 0$ of the analytic function $\Theta_Q(z)$ in the half plane $\text{Re } z > 0$. The main ingredient in the proof is then the theorem that if the boundary value v (in distribution sense) of an analytic function u in a half plane is smooth in a neighborhood of the boundary point z_0 , then u is smooth up to the boundary in a neighborhood of z_0 .

If P is a differential operator then no logarithmic terms occur in the asymptotic expansion (2.15), corresponding to the fact that $\Gamma(s)Z_p(s)$ only has simple poles. However, for general pseudo-differential operators P poles of $Z_p(s)$ can occur at any point $(n-k)/m$, $k \neq n$. Indeed, replacing Q by $Q+c$, $c > 0$ amounts to replacing μ by $\mu-c$ in the asymptotic expansion (2.1), so for a suitable choice of c we have $\omega_k(x) \neq 0$ for any $k \neq n$ for the new operator if we had $\omega_{n+1}(x) \neq 0$ for Q . Now according to (2.7) $\omega_{n+1}(x)$ is an integral over $S_x^* X$ of $i \partial a_{-n} / \partial t(0, x, \xi)$, which can be made arbitrarily large by a sufficiently large choice of $Q_{-n}(x, \xi)$, keeping the homogeneous terms of degree $> -n$ in the symbol of Q fixed.

Integrating (2.1) over X leads to

$$\sum_j \rho(\mu - \mu_j) \sim (2\pi)^{-n} \sum_{k=0}^{\infty} c_k \mu^{n-1-k} \quad \text{for } \mu \rightarrow \infty, \quad (2.16)$$

and rapidly decreasing as $\mu \rightarrow -\infty$.

Here

$$c_k = \int_X \omega_k, \quad (2.17)$$

so in particular

$$c_0 = \text{vol}(S^*X), \quad c_1 = (1-n) \int_{S^*X} \text{sub } Q. \quad (2.18)$$

In view of Corollary 1.2, the asymptotic expansion (2.16) does not change if we replace ρ by any $\rho \in \mathcal{S}(\mathbb{R})$ such that $\hat{\rho}$ has compact support containing no period $T \neq 0$ of a periodic H_q solution curve, and all derivatives of $1 - \hat{\rho}$ at $t=0$ vanish.

Integrating (2.13), resp. (2.15) over X leads to the analogue of Corollary 2.2, resp. 2.2' for the ζ -function $\zeta_P(s) = \sum_j \lambda_j^{-s}$, resp. the “heat function” $\mathfrak{H}_P(z) = \sum_j e^{-\lambda_j z}$.

We conclude this Section by some basic estimates for the eigenvalues of Q which follow from a consideration of only the top order terms in (2.16).

Lemma 2.3. *For any $K > 0$ we have*

$$\# \{j; |\mu_j - \mu| \leq K\} = O(\mu^{n-1}) \quad \text{as } \mu \rightarrow \infty. \quad (2.19)$$

Proof. Choose $\rho \in \mathcal{S}(\mathbb{R})$ such that $\hat{\rho}$ has compact support, $\int \hat{\rho}(t) dt \neq 0$. Replacing ρ by $\rho \cdot \bar{\rho}$ one obtains that $\rho \geq 0$, $\rho(0) > 0$ and still $\hat{\rho}$ having compact support. Replacing $\hat{\rho}$ by $t \mapsto \hat{\rho}(t/\delta)$ amounts to replacing ρ by $\mu \mapsto \delta \cdot \rho(\mu\delta)$. Taking δ small one gets $\text{supp } \hat{\rho}$ in any prescribed neighborhood of 0 and $\rho > 0$ on $[-K, K]$. An examination of the first part of the proof of Proposition 2.1 shows that the condition that $\hat{\rho} = 1$ on a neighborhood of 0 is not necessary in order to obtain an asymptotic expansion of the for (2.16), possibly with other constants c_k . Now (2.19) follows from

$$\# \{j; |\mu_j - \mu| \leq K\} \cdot \min \{\rho(\mu); \mu \in [-K, K]\} \leq \sum_j \rho(\mu - \mu_j) = O(\mu^{n-1}).$$

Lemma 2.4. *For every $\rho \in \mathcal{S}(\mathbb{R})$ and every $\varepsilon > 0$ there exists a number K such that*

$$\sum_{\{j; |\mu_j - \mu| > K\}} \rho(\mu - \mu_j) \leq \varepsilon \cdot \mu^{n-1}, \quad (2.20)$$

$$\sum_{\{j; \mu_j > v + K\}} \left| \int_{-\infty}^v \rho(\mu - \mu_j) d\mu \right| \leq \varepsilon v^{n-1}, \quad (2.21)$$

and

$$\sum_{\{j; \mu_j < v - K\}} \left| \int_v^{\infty} \rho(\mu - \mu_j) d\mu \right| \leq \varepsilon v^{n-1} \quad (2.22)$$

for all $\mu, v \geq 1$.

Proof. For every N there is a C_N such that $|\rho(\mu)| \leq C_N(1 + |\mu|)^{-N}$ for all μ . On the other hand (2.19) implies the existence of a constant C such that

$$\# \{j; |\mu_j - \mu| \leq \tfrac{1}{2}\} \leq C(\tfrac{1}{2} + |\mu|)^{n-1} \quad \text{for all } \mu.$$

Combining these two estimates one obtains for $\mu \geq 0$, $N > n$:

$$\begin{aligned} \sum_{|\mu_j - \mu| > K} |\rho(\mu - \mu_j)| &= \sum_{k=0}^{\infty} \sum_{K+k+1 \leq |\mu_j - \mu| \leq K+k} |\rho(\mu - \mu_j)| \\ &\leq 2 C C_N \sum_{l=K+1}^{\infty} (\mu + l)^{n-1} l^{-N} \\ &\leq 2 C C_N \sum_{j=0}^{n-1} \binom{n-1}{j} \mu^{n-1-j} K^{j-N+1} / (N-j-1). \end{aligned}$$

So (2.20) follows by taking K sufficiently large. (2.21), resp. (2.22) follow by writing

$$\int_{-\infty}^v \rho(\mu - \mu_j) d\mu = \int_{-\infty}^{v-\mu_j} \rho(\mu) d\mu,$$

resp.

$$\int_v^{\infty} \rho(\mu - \mu_j) d\mu = \int_{v-\mu_j}^{\infty} \rho(\mu) d\mu$$

and using the rapid decrease of $\sigma \mapsto \int_{-\infty}^{\sigma} \rho(\mu) d\mu$, resp. $\sigma \mapsto \int_{\sigma}^{\infty} \rho(\mu) d\mu$ as $\sigma \rightarrow -\infty$, resp. $\sigma \rightarrow \infty$.

Corollary 2.5.

$$\# \{j; \mu_j \leq v\} = (2\pi)^{-n} \cdot \text{vol}(B^* X) \cdot v^n + O(v^{n-1}) \quad \text{as } v \rightarrow \infty. \quad (2.23)$$

Proof. Integrating (2.16) from $-\infty$ to v one obtains

$$\int_{-\infty}^v \sum_{j=1}^{\infty} \rho(\mu - \mu_j) d\mu = (2\pi)^{-n} \sum_{k=0}^{n-1} c_k v^{n-k} / (n-k) + O(1) \quad (2.24)$$

for $v \rightarrow \infty$. On the other hand

$$\begin{aligned} \int_{-\infty}^v \sum_{j=1}^{\infty} \rho(\mu - \mu_j) d\mu &= \sum_{\{j; \mu_j > v+K\}} \int_{-\infty}^v \rho(\mu - \mu_j) d\mu \\ &+ \sum_{\{j; |v-\mu_j| \leq K\}} \int_{-\infty}^v \rho(\mu - \mu_j) d\mu + \sum_{\{j; \mu_j < v-K\}} \int_{-\infty}^{\infty} \rho(\mu - \mu_j) d\mu \\ &- \sum_{\{j; \mu_j < v-K\}} \int_v^{\infty} \rho(\mu - \mu_j) d\mu. \end{aligned}$$

Now

$$\int_{-\infty}^{\infty} \rho(\mu - \mu_j) d\mu = 1$$

and the other terms are all $O(v^{n-1})$ as $v \rightarrow \infty$ in view of (2.21), (2.19) and (2.22), so

$$\# \{j; \mu_j < v-K\} = (2\pi)^{-n} c_0 v^n / n + O(v^{n-1}) \quad \text{for } v \rightarrow \infty.$$

Replacing v by $v+K$ in this formula gives (2.23), using that

$$\text{vol} \{(x, \xi) \in T^* X; q(x, \xi) \leq 1\} = \text{vol}(S^* X) \int_0^1 \omega^{n-1} d\omega = c_0 / n.$$

Remarks. Applying the same argument to (2.1) leads to

$$\sum_{\mu_j \leq v} |e_j(x)|^2 = (2\pi)^{-n} \cdot \text{vol}(B_x^* X) \cdot v^n + O(v^{n-1}), \quad v \rightarrow \infty \quad (2.25)$$

uniformly in x , here $B_x^* X = \{\xi \in (T_x^* X)^*; q(x, \xi) \leq 1\}$. This is the main result of Hörmander [16]. Of course (2.23) follows from (2.25) by integration over X .

We also remark that (2.16), (2.20) easily imply that there exist $K > 0$, $\delta > 0$, v such that

$$\# \{j; |\mu_j - \mu| \leq K\} \geq \delta \cdot \mu^{n-1} \quad \text{for all } \mu \geq v. \quad (2.26)$$

3. Periodic H_q Flow

Theorem 3.1. *Suppose that all H_q -solution curves are periodic with period $T > 0$ and that the average of sub Q over these curves is equal to a constant γ . Write*

$$v_k = \frac{2\pi}{T} (k + \frac{1}{4}\alpha) + \gamma, \quad k = 1, 2, \dots, \quad (3.1)$$

here α is the index of the $\partial/\partial t + H_q$ solution curves of period T given by the Arnol'd cohomology class in $H^1(\tilde{C}, \mathbb{Z})$ (see Arnol'd [2], Hörmander [17, (3.2.15)]), $\tilde{C}$ is the projection into $T^*((\mathbb{R} \bmod T) \times X \times X)$ of the canonical relation C defined in (1.15).

Then there exist positive constants $K, K', \tilde{\mu}$ such that

$$\begin{aligned} \# \{j; |\mu_j - \mu| \leq K, |\mu_j - v_k| \leq Bk^{-\frac{1}{2}} \text{ for some } k \in \mathbb{N}\} \\ \geq (1 - K'/B^2) \cdot \# \{j; |\mu_j - \mu| \leq K\} \quad \text{for all } B > 0, \mu \geq \tilde{\mu}. \end{aligned} \quad (3.2)$$

Moreover, if $\sup Q = 0$ then $\gamma = 0$ and the number $Bk^{-\frac{1}{2}}$ in the left hand side of (3.2) can be replaced by Bk^{-1} .

Proof. $\Phi^{t+T} = \Phi^t$ for all $t \in T$ implies that $U(t+T, x, y)$ is defined by the same canonical relation C as $U(t, x, y)$. The principal symbol u of U satisfies the transport equation

$$i^{-1} \mathcal{L}_{(\partial/\partial t + H_q)} u + (\text{sub } Q) \cdot u = 0 \quad (3.3)$$

([9, Theorem 5.3.1]), and it follows that the principal symbol of $U(t+T, x, y)$ is equal to $i^{-\alpha} e^{-i\gamma T} u$. Indeed, noting that γT is the integral of sub Q over the $\partial/\partial t + H_q$ solution curve, t running from 0 to T , we see, in the notation of [17, Section 3.2], that

$$s_j = i^{\sigma_{j,j-1} + \dots + \sigma_{1,0}} e^{-i\gamma T} s_0,$$

whereas on the other hand

$$\alpha = \sigma_{0,1} + \dots + \sigma_{j-1,j} = -(\sigma_{j,j-1} + \dots + \sigma_{1,0})$$

if the φ_k , $k = 0, \dots, j$ are phase functions defining $\tilde{C}'$ on patches along a $\partial/\partial t + H_q$ solution curve of period T . Writing $\frac{1}{2}\pi\alpha + \gamma T = \beta T$ it follows that

$$U(t, x, y) - e^{i\beta T} U(t+T, x, y)$$

is a Fourier integral operator defined by C of one lower order $-\frac{1}{4} - 1$, so the integral

$$(2\pi)^{-1} \int e^{i\mu t} \hat{\rho}(t) [\hat{\sigma}(t) - e^{i\beta T} \hat{\sigma}(t+T)] dt \quad (3.4)$$

can be written in the form (1.25) with the amplitude b now of order -1 , and we see that (3.4) is of order $O(\mu^{n-2})$. Choosing $\rho \geq 0$ and taking real parts in (3.4) gives

$$\sum_j (1 - \cos((\beta - \mu_j) T)) \cdot \rho(\mu - \mu_j) \leq A \mu^{n-2}, \quad \mu \geq 1, \quad (3.5)$$

for some constant A . Here it is used that $t \mapsto \hat{\sigma}(t + T)$ is equal to the Fourier transform of

$$\sum_j e^{-i\mu_j T} \delta(\mu - \mu_j).$$

Now define for each $B > 0$, $k \in \mathbb{N}$:

$$N(k, B) = \# \{j; B k^{-\frac{1}{2}} \leq T|\mu_j - v_k| \leq \pi\}. \quad (3.6)$$

Using that $1 - \cos(2\pi k + x) \geq \frac{1}{2} x^2$ for $|x| \leq \pi$ and substituting $\mu = v_k$ in (3.5) one obtains the estimate

$$N(k, B) \cdot \frac{1}{2} \cdot B^2 k^{-1} \cdot \min\{\rho(\mu); |\mu| \leq \pi/T\} \leq A' k^{n-2} \quad (3.7)$$

for all $k \in \mathbb{N}$ and some A' . Taking ρ such that $\rho(\mu) > 0$ for $|\mu| \leq \pi/T$ (see the proof of Lemma 2.3) this leads to $N(k, B) \leq A'' B^{-2} k^{n-1}$ for all $B > 0$, $k \in \mathbb{N}$ and still another constant A'' . Combining this with (2.26) the estimate (3.2) readily follows.

If $\text{sub } Q = 0$ then it follows from the self-adjointness of Q that the homogeneous part $Q_0(x, \xi)$, resp. $Q_{-1}(x, \xi)$ of order 0, resp. -1 of the total symbol of Q on local coordinates is purely imaginary, resp. real. Representing $U(t, x, y)$ locally by integrals (1.17) with $\frac{\partial \varphi}{\partial t}(t, x, y, \zeta) + q(x, d_x \varphi(t, x, y, \zeta))$ being identically zero

(instead of only on $d_\zeta \varphi(t, x, y, \zeta) = 0$), as can always be done, one obtains for $(i^{-1} \partial/\partial t + Q) \circ U$ a similar local representation with a replaced by some amplitude $\tilde{a}$, again of order 0. It turns out that $\tilde{a}_0 = i^{-1} L a_0$, with L a real first order linear partial differential operator. Because we can choose $a_0(0, x, y, \zeta) = 1$ as we have seen in the proof of Proposition 2.1 it follows that locally solving $\tilde{a}_0 = 0$ leads to an a_0 which on the k -th patch along the $\partial/\partial t + H_q$ solution curve is equal to $i^{\sigma_k, k-1+\dots+\sigma_1, 0}$ times a real function. With this choice of the a_0 , and using that iQ_0, Q_{-1} are real, one obtains $\tilde{a}_{-1} = i^{-1} L a_{-1} + f$, with $i^{-(\sigma_k, k-1+\dots+\sigma_1, 0)} \cdot f$ real-valued. Because we have seen in the proof of Proposition 2.1 that $a_{-1}(0, x, y, \zeta)$ can be chosen purely imaginary, solving $\tilde{a}_{-1} = 0$ locally leads to an a_{-1} such that on the k -th patch $i^{-(\sigma_k, k-1+\dots+\sigma_1, 0)} a_{-1}$ is purely imaginary. It follows that in the representation of (3.4) we may assume that b_{-1} is purely imaginary. However, taking real parts in (3.4) then shows that in (3.5) we may replace μ^{n-2} by μ^{n-3} and the argument goes through with $k^{-\frac{1}{2}}$ replaced by k^{-1} in the definition of $N(k, B)$. This proves the last statement in the theorem. (Of course the proof would have been much cleaner if we would have had an invariant analogue of [9, Theorem 5.3.1] for a subprincipal symbol of the product.)

The estimate (3.2) means that (by choosing B sufficiently large) an arbitrary large fraction of the eigenvalues in long and far away intervals lies in the intervals around the v_k with decreasing radius $B \cdot k^{-\frac{1}{2}}$ ($B \cdot k^{-1}$ if $\text{sub } Q = 0$). There is also a converse to Theorem 3.1.

Theorem 3.2. *Let X be connected and $n \geq 2$. Suppose that there is a constant K such that for each $\varepsilon > 0$ there is a number $\tilde{\mu}$ such that*

$$\begin{aligned} \# \{j; |\mu_j - \mu| \leq K, |\mu_j - \nu_k| \leq \varepsilon \text{ for some } k \in \mathbb{N}\} \\ \geq (1 - \varepsilon) \cdot \# \{j; |\mu_j - \mu| \leq K\} \text{ for all } \mu \geq \tilde{\mu}. \end{aligned} \quad (3.8)$$

Here $\nu_k = 2\pi k/T + \beta$, $k \in \mathbb{N}$, β a fixed real number. Then all H_q solution curves are periodic with period T , the $0 - T$ -average of $\text{sub } Q$ over these curves is equal to a constant γ , $\beta = \frac{1}{2}\pi\alpha/T + \gamma$ modulo $2\pi/T$, and (3.8) can be replaced by the stronger estimates of Theorem 3.1.

Proof. (3.8) implies that

$$\sum_j (1 - e^{i(\beta - \mu_j)T}) \cdot \rho(\mu - \mu_j) = o(\mu^{n-1}) \quad \text{for } \mu \rightarrow \infty. \quad (3.9)$$

For the proof, split (3.9) as a sum over the j such that $|\mu - \mu_j| \leq K$, $|\mu_j - \nu_k| \geq \varepsilon$ for all $k \in \mathbb{N}$, a sum over the j such that $|\mu_j - \nu_k| < \varepsilon$ for some $k \in \mathbb{N}$, and the rest. Each of the summands can be estimated from above by $\varepsilon' \cdot \mu^{n-1}$ for arbitrary $\varepsilon' > 0$, the last one by applying (2.20) and choosing K sufficiently large, the second one by observing that $1 - \exp i(\beta - \mu_j)T$ becomes small and using that $\sum_j \rho(\mu - \mu_j) = O(\mu^{n-1})$, and finally the first one by combining (3.8) with (2.19), taking ε sufficiently small and $\mu \geq \tilde{\mu}$, $\tilde{\mu}$ sufficiently large.

(3.9) can also be written as

$$(2\pi)^{-1} \int e^{i\mu t} \hat{\rho}(t) [\hat{\sigma}(t) - e^{i\beta T} \hat{\sigma}(t+T)] dt = o(\mu^{n-1}) \quad (3.10)$$

for $\mu \rightarrow \infty$. $\hat{\rho}(t) e^{i\beta T} U(t+T, x, y)$ can be written as a finite sum of integrals of the form (1.17) with phase function $\varphi = \varphi^r$ and amplitude $a = a^r$ in the r -th summand. Using only the top order term in (2.16) it follows as in the beginning of the proof of Proposition 2.1 that

$$\begin{aligned} \sum_r \int_{q(x, \tilde{\zeta})=1} e^{i\mu t^r(x, \tilde{\zeta})} a_0^r(t^r(x, \tilde{\zeta}), x, x, \tilde{\zeta}) dx d\tilde{\zeta} \\ = \text{vol}(S^*X) + o(1) \quad \text{for } \mu \rightarrow \infty. \end{aligned} \quad (3.11)$$

Here $t^r(x, \tilde{\zeta})$ is equal to the local solution t of the equation $\varphi^r(t, x, x, \tilde{\zeta}) = 0$ and a_0^r is the zero order part of a^r .

Using the principle of stationary phase the only contributions to (3.11) which are not $o(1)$ for $\mu \rightarrow \infty$ come from the points $(x, \tilde{\zeta})$ where all derivatives of t^r of order ≥ 1 vanish. Because the estimate (3.11) holds for $\text{supp } \hat{\rho}$ contained in an arbitrarily small neighborhood of 0 it follows that

$$\sum_r \int_{\Omega^r} a_0^r(0, x, x, \tilde{\zeta}) dx d\tilde{\zeta} = \text{vol}(S^*X), \quad (3.12)$$

here Ω^r is the set of $(x, \tilde{\zeta}) \in S^*X$ such that t^r vanishes of infinite order at $(x, \tilde{\zeta})$. From (1.18), (1.15) it follows that Φ^T has a contact of infinite order with the identity at $(x, d_x \varphi^r(0, x, x, \tilde{\zeta}))$ such that $(x, \tilde{\zeta}) \in \Omega^r$, and in view of the proof of Theorem 3.1.6 in [17] we see that φ^r can be chosen such that it has arbitrary order of contact with the phase function φ in (2.5) at the points $(0, x, x, \tilde{\zeta})$ such that $(x, \tilde{\zeta}) \in \Omega^r$. In fact, the contributions to (3.11) which are not $o(1)$ as $\mu \rightarrow \infty$ can be represented

by only one integral with such a phase function φ^r , for which we then have

$$\int_{\Omega^r} a_0^r(0, x, x, \tilde{\zeta}) dx d\tilde{\zeta} = \text{vol}(S^*X). \quad (3.13)$$

Because with the phase function φ of (2.5) the amplitude in the integral representation of $U(0, x, x)$ has top order term equal to 1, it follows in view of the transport Eq. (3.3) for the principal symbol of U that

$$a_0^r(0, x, x, \tilde{\zeta}) = i^{-\alpha} e^{-i\gamma(x, \tilde{\zeta}) \cdot T} \quad \text{for } (x, \tilde{\zeta}) \in \Omega^r, \quad (3.14)$$

here $\gamma(x, \tilde{\zeta}) \cdot T$ is equal to the $0 - T$ -integral of sub Q over the H_q solution curve starting and ending at $(x, \tilde{\zeta}) \in \Omega^r$. It follows that (3.13) can only be possible if $\frac{1}{2} \pi i \alpha + \gamma(x, \tilde{\zeta}) \cdot T$ is equal to 0 modulo 2π for almost all $(x, \tilde{\zeta}) \in \Omega^r$, and moreover $S^*X \setminus \Omega^r$ has $(2n-1)$ -dimensional measure equal to 0. Because Ω^r is closed, $\Omega^r = S^*X$, that is $\Phi^T = I$. Because γ is continuous and S^*X is connected (if X is connected and $n > 1$) it follows that γ is constant (note that α is locally constant by definition) and the theorem is proved.

If we make some additional assumptions on the H_q flow (excluding some rather pathological situations) the above results can be improved considerably. The idea is that if we want a sharp estimate for $\#\{j; |\mu_j - \mu| \leq \frac{1}{2}\varepsilon\}$ from $\sum_j \rho(\mu - \mu_j)$, then it is better to let ρ resemble the step function

$$\chi_\varepsilon(\mu) = \begin{cases} 1/\varepsilon & \text{for } |\mu| \leq \frac{1}{2}\varepsilon, \\ 0 & \text{elsewhere.} \end{cases} \quad (3.15)$$

$\hat{\chi}_\varepsilon$ is a smooth function with $\hat{\chi}_\varepsilon(0) = 1$, but it has no compact support, so we multiply it by $\hat{\rho}_\delta: t \mapsto \hat{\rho}(t \cdot \delta)$, with $\rho \geq 0$, $\hat{\rho} \in C_0^\infty(\mathbb{R})$, $\hat{\rho}(0) = 1$, in order to cut it off. On the inverse Fourier transform side this amounts to taking the convolution $\rho_{\varepsilon, \delta} = \chi_\varepsilon * \rho_\delta$. Of course $\rho_{\varepsilon, \delta} \in \mathcal{S}(\mathbb{R})$ and for $\delta \rightarrow 0$ it converges to χ_ε . In fact:

Lemma 3.3. *For every N there is a constant $M > 0$ such that*

$$\begin{aligned} \varepsilon^{-1} \cdot \#\{j; |\mu_j - \mu| \leq \tfrac{1}{2}(\varepsilon - \varepsilon')\} \cdot (1 - (\delta/\varepsilon')^N) \\ \leq \sum_j \rho_{\varepsilon, \delta}(\mu - \mu_j) \\ \leq \varepsilon^{-1} \cdot \#\{j; |\mu_j - \mu| \leq \tfrac{1}{2}(\varepsilon + \varepsilon')\} + (\delta/\varepsilon')^N \cdot \mu^{n-1} \end{aligned} \quad (3.16)$$

for all $\mu, \delta, \varepsilon, \varepsilon'$ such that $\mu \geq 1 \geq \varepsilon \geq \varepsilon' \geq M\delta > 0$.

Proof. Note that

$$\rho_{\varepsilon, \delta}(v) = \varepsilon^{-1} \cdot \int_{(v - \frac{1}{2}\varepsilon)/\delta}^{(v + \frac{1}{2}\varepsilon)/\delta} \rho(\mu) d\mu.$$

For the first inequality use that $\rho_{\varepsilon, \delta} \geq 0$ and that $1 - \int_{-\infty}^v \rho(\mu) d\mu$ is rapidly decreasing as $v \rightarrow \infty$. For the second inequality the summation is split up into one over the j such that $|\mu_j - \mu| \leq \frac{1}{2}(\varepsilon + \varepsilon')$ and the rest into sums over $\mu - \frac{1}{2}(\varepsilon + \varepsilon') - (k+1) \leq \mu_j \leq \mu - \frac{1}{2}(\varepsilon + \varepsilon') - k$, resp. $\mu + \frac{1}{2}(\varepsilon + \varepsilon') + k \leq \mu_j \leq \mu + \frac{1}{2}(\varepsilon + \varepsilon') + (k+1)$. The latter sums are estimated using (2.19) and the rapid decrease of $\int_{-\infty}^v \rho(\mu) d\mu$ and $\int_v^\infty \rho(\mu) d\mu$ as $v \rightarrow \infty$, and then the result is summed over $k = 0, 1, 2, \dots$

As a first application of the use of these cut-off functions we improve the estimates in Theorem 3.1. Note that we allowed there a big constant K , which makes it possible than in fact many of the intervals $|\mu_j - v_k| \leq B k^{-\frac{1}{2}}$ are quite empty. This would happen for instance if the H_q flow is periodic with period $T_0 > 0$ and $T = l \cdot T_0$ for some integer $l > 1$. Then the majority of the eigenvalues will only cluster around the v_k for $k = l, 2l, 3l, \dots$ and not around the other v_k . Now let T be the minimal positive common period of all H_q solution curves. Then a solution curve is called *subperiodic* if it is periodic with a positive period which is not an integer multiple of T , it is then automatically periodic with a period T/l for some integer $l > 0$.

Theorem 3.4. *Assume that the H_q flow is periodic with minimal positive period T and that the union of the subperiodic orbits has measure zero in $T^*X \setminus 0$. Assume also that the average of sub Q over the $0 - T$ -solution curves of the H_q flow is equal to a constant γ . Then there exist K', k_0 such that*

$$\# \left\{ j; \frac{\pi}{T} \geq |\mu_j - v_k| > B k^{-\frac{1}{2}} \right\} \leq K' \cdot B^{-2} v_k^{n-1} \quad (3.17)$$

for all $B > 0, k \geq k_0$. On the other hand

$$\lim_{B, k \rightarrow \infty} \# \{ j; |\mu_j - v_k| \leq B k^{-\frac{1}{2}} \} / v_k^{n-1} = T^{-1} (2\pi)^{1-n} \text{vol}(S^*X). \quad (3.18)$$

Here the v_k are as in (3.1) and again we may replace $B \cdot k^{-\frac{1}{2}}$ by $B \cdot k^{-1}$ in the left hand sides of (3.17) and (3.18) if sub $Q = 0$.

Proof. The set of $(x, \xi) \in S^*X$ such that $\Phi^t(x, \xi) = (x, \xi)$ for some t which is not an integer multiple of T has measure 0 in S^*X . Using computations as in the beginning of the proof of Proposition 2.1 it follows that the contribution to (1.24) coming from such periods of H_q solution curves is of order $o(\mu^{n-1})$ as $\mu \rightarrow \infty$. So modulo terms of order $o(\mu^{n-1})$ we get only contributions from neighborhoods of $t = l \cdot T, l \in \mathbb{Z}$, and iterating (3.10) and using the estimate (2.16) for the contribution from the singularity at $t = 0$, it follows that

$$\sum_j \rho(\mu - \mu_j) = \sum_{l \in \mathbb{Z}} e^{i(\mu - \beta)lT} \cdot \hat{\rho}(lT) \cdot (2\pi)^{-n} \cdot c_0 \cdot \mu^{n-1} + o(\mu^{n-1}) \quad \text{as } \mu \rightarrow \infty \quad (3.19)$$

for every $\hat{\rho} \in C_0^\infty(\mathbb{R})$. (Note that the sum in the right hand side of (3.19) is finite.)

Substituting $\rho = \rho_{\varepsilon, \varepsilon\delta}, \mu = v_k$ and using that

$$\begin{aligned} \sum_{l \in \mathbb{Z}} \hat{\chi}_\varepsilon(lT) \hat{\rho}_{\varepsilon\delta}(lT) &= (\varepsilon T)^{-1} \cdot \varepsilon T \sum_{l \in \mathbb{Z}} \hat{\chi}_1(l \cdot \varepsilon T) \cdot \hat{\rho}_\delta(l \cdot \varepsilon T) \\ &= (\varepsilon T)^{-1} \int \hat{\chi}_1(t) \cdot \hat{\rho}_\delta(t) dt + O_\delta(1) \quad \text{as } \varepsilon \searrow 0 \end{aligned}$$

and

$$\int \hat{\chi}_1(t) \hat{\rho}_\delta(t) dt \rightarrow 2\pi \quad \text{as } \delta \searrow 0,$$

it follows from Lemma 3.3 that for every $\varepsilon > 0$ one has

$$\# \{ j; |\mu_j - v_k| \leq \varepsilon \} = \frac{2\pi}{T} \cdot (2\pi)^{-n} c_0 v_k^{n-1} + o(v_k^{n-1}) \quad (3.20)$$

for $k \rightarrow \infty$. Using (3.2) this implies (3.17) and using (3.17), (3.20) can be improved to (3.18).

Remark. Weinstein [24] obtained a very precise estimate about the resemblance between the spectra of operators P, P' such that the principal symbol of P' is given by $p \circ C$, C a homogeneous canonical transformation, p = principal symbol of P . For a Zoll surface this leads to the conclusion that all the eigenvalues of the Laplacian lie in the union of intervals of the form $[j^2 - j - B, j^2 - j + B]$, $j \in \mathbb{N}$, and the number of eigenvalues in such an interval is equal to $j^2 - j$ for sufficiently large j . Comparing this with (3.17), (3.18) for $Q = (-\Delta + c)^{\frac{1}{2}}$ (sub $Q = 0$ in this case), we see that we have a similar estimate for the intervals in which the majority of the eigenvalues lie, but that Weinstein's results for both the number of eigenvalues inside these intervals and those outside are much more precise. (On the other hand our estimates are valid for more general operators.)

Theorem 3.5. *Suppose that the set of $(x, \xi) \in S^*X$ such that Φ^t has a contact of infinite order with the identity at (x, ξ) for some $t \neq 0$, has zero measure in S^*X . Then for every $\varepsilon > 0$:*

$$\lim_{\mu \rightarrow \infty} \# \{j; |\mu_j - \mu| \leq \tfrac{1}{2}\varepsilon\} / \mu^{n-1} = \varepsilon \cdot (2\pi)^{-n} \text{vol}(S^*X). \quad (3.21)$$

Furthermore,

$$\begin{aligned} \# \{j; \mu_j \leq v\} &= (2\pi)^{-n} \cdot \text{vol}(B^*X) \cdot v^n \\ &\quad - (2\pi)^{-n} \cdot \left(\int_{S^*X} \text{sub } Q \right) \cdot v^{n-1} + o(v^{n-1}) \quad \text{as } v \rightarrow \infty, \end{aligned} \quad (3.22)$$

improving (2.23) just a tiny little bit.

Proof. The assumption implies that all contributions to (1.24) coming from t in a neighborhood of periods $T \neq 0$ are of order $o(\mu^{n-1})$ as $\mu \rightarrow \infty$. Substituting $\rho = \rho_{\varepsilon, \varepsilon\delta}$, using (2.16) and (3.16) and letting $\delta \searrow 0$, (3.21) readily follows.

Secondly, if $\hat{\alpha} \in C_0^\infty(\mathbb{R})$, $0 \notin \text{supp } \hat{\alpha}$, then

$$\begin{aligned} \int_{-\infty}^v \sum_j \rho(\mu - \mu_j) d\mu &= \lim_{a \rightarrow -\infty} (2\pi)^{-1} \int_a^v e^{i\mu t} \hat{\alpha}(t) \hat{\sigma}(t) dt d\mu \\ &= (2\pi)^{-1} \int e^{ivt} (it)^{-1} \hat{\alpha}(t) \hat{\sigma}(t) dt \\ &\quad - \lim_{a \rightarrow -\infty} (2\pi)^{-1} \cdot \int e^{iat} (it)^{-1} \hat{\alpha}(t) \hat{\sigma}(t) dt \\ &= (2\pi)^{-1} \int e^{ivt} (it)^{-1} \hat{\alpha}(t) \hat{\sigma}(t) dt \\ &= o(v^{n-1}) \quad \text{as } v \rightarrow \infty. \end{aligned}$$

So replacing ρ by ρ_δ in (2.24) leads to

$$\begin{aligned} \int_{-\infty}^v \sum_j \rho_\delta(\mu - \mu_j) d\mu &= (2\pi)^{-n} c_0 \cdot v^n / n + (2\pi)^{-n} c_1 \cdot v^{n-1} / (n-1) \\ &\quad + o_\delta(v^{n-1}) \quad \text{as } v \rightarrow \infty. \end{aligned}$$

Note that the assumption implies that $n > 1$.

Now, for any $\varepsilon > 0$ we can obtain both

$$\left| \sum_{\{j; |v - \mu_j| \leq K\}} \int_{-\infty}^v \rho_\delta(\mu - \mu_j) d\mu \right| \leq \# \{j; |\mu_j - v| \leq K\} \cdot \int_{-\infty}^{\infty} |\rho_\delta(\mu)| d\mu \leq \varepsilon \cdot v^{n-1}$$

and

$$|\# \{j; \mu_j < v - K\} - \# \{j; \mu_j \leq v\}| = \# \{j; v - K \leq \mu_j \leq v\} \leq \varepsilon \cdot v^{n-1}$$

by choosing $K > 0$ sufficiently small (independent of δ) and v sufficiently large, using (3.21).

With this choice of K , the quantity

$$\left| \sum_{\{j; \mu_j > v + K\}} \int_{-\infty}^v \rho((\mu - \mu_j)/\delta) d\mu/\delta \right|$$

can be estimated from above by a constant times

$$\int_{v+K}^{\infty} \omega^{n-1} \int_{-\infty}^{(v-\omega)/\delta} |\rho(\mu)| d\mu d\omega = \int_{-\infty}^{-K/\delta} \frac{1}{n} ((v - \delta\mu)^n - (v + K)^n) \cdot |\rho(\mu)| d\mu,$$

and therefore is smaller than $\varepsilon \cdot v^{n-1}$ if $\delta > 0$ is sufficiently small and v sufficiently large. Similarly

$$\left| \sum_{\{j; \mu_j < v - K\}} \int_v^{\infty} \rho_\delta(\mu - \mu_j) d\mu \right| \leq \varepsilon v^{n-1}$$

if $\delta > 0$ is sufficiently small and v sufficiently large.

Combining these estimates gives

$$\left| \int_{-\infty}^v \sum_j \rho_\delta(\mu - \mu_j) d\mu - \# \{j; \mu_j \leq v\} \right| \leq 4\varepsilon \cdot v^{n-1}$$

for $\delta > 0$ sufficiently small and v sufficiently large, completing the proof of (3.22).

(3.21), (3.22) express that for large v the eigenvalues of Q are quite evenly distributed on the real axis, in sharp contrast with the clustering described in Theorem 3.1, 3.4. If X is a real-analytic manifold and q is a real analytic function on $T^*X \setminus 0$, then the assumption in Theorem 3.5 holds as soon as there is no $T \neq 0$ such that all H_q solution curves are periodic with period T .

Question. What happens (in terms of estimates for finite μ_j -intervals) if there is a $T \neq 0$ such that all H_q solution curves are periodic with period T , but the average of sub Q over these curves is not constant?

4. The Singularities of $\hat{\sigma}(t)$ for $t \neq 0$

We recall that the singularities of $\hat{\sigma}$ occur at the set of periods of the periodic H_q solution curves (Corollary 1.2). If the H_q solution curves of period T form a “nice” submanifold of S^*X we can obtain more precise information on the singularity at T . We first need a definition (due to Bott [5]).

Definition 4.1. Let M be a manifold and let $\Phi: M \rightarrow M$ be a diffeomorphism. A submanifold $Z \subset M$ of fixed points of Φ is called *clean* if for each $z \in Z$ the set of fixed points of $d\Phi_z: T_z M \rightarrow T_z M$ equals the tangent space to Z at z .

We will show that if M and Φ are symplectic, Z possesses an intrinsic positive measure. This depends on the following three elementary lemmas.

Lemma 4.2. *Let V be a symplectic vector space with two-form Ω and let $P: V \rightarrow V$ be a symplectic linear mapping. Then $\ker(I - P)$ and $\operatorname{coker}(I - P)$ are canonically paired by Ω .*

Proof. If $v \in \ker(I - P)$ then $v = Pv$, so $v \in \ker(I - P^{-1})$. This implies $\Omega(v, (I - P)w) = 0$ for all w , or that $v \in \operatorname{im}(I - P)^\perp$. Hence $\ker(I - P)$ and $\operatorname{coker}(I - P)$ are canonically paired. Q.E.D.

Now consider the exact sequence

$$0 \rightarrow \ker \rightarrow V \xrightarrow{I-P} V \rightarrow \operatorname{coker} \rightarrow 0. \quad (4.1)$$

Letting $|\cdot|^\alpha$ be the functor that assigns to each vector space V its one-dimensional subspace of α -densities,

$$|\ker|^\frac{1}{2} \otimes |V|^{-\frac{1}{2}} \otimes |V|^\frac{1}{2} \otimes |\operatorname{coker}|^{-\frac{1}{2}} \cong 1. \quad (4.2)$$

Since $|V|^\frac{1}{2} \otimes |V|^{-\frac{1}{2}} \cong 1$ and $|\operatorname{coker}|^\frac{1}{2} \cong |\ker|^{-\frac{1}{2}}$ by the lemma, we get $|\ker| \cong 1$; so we've proved:

Lemma 4.3. *If P is a symplectic mapping of V onto V then $\ker(I - P)$ possesses a canonical density.*

From $I - P$ we get a linear map

$$(I - P)^*: V/\ker \rightarrow V/\ker, \quad \ker = \ker(I - P). \quad (4.3)$$

If (4.3) is onto, then

$$V = \ker(I - P) \oplus \operatorname{Im}(I - P). \quad (4.4)$$

Moreover, by Lemma 4.2, Ω restricted to both factors is non-degenerate. Substituting the left hand side of (4.4) for V in (4.2) and making obvious cancellations we get

Lemma 4.4. *If (4.3) is onto, the canonical density just defined on $\ker(I - P)$ is equal to*

$$|\det(I - P)^*|^{-\frac{1}{2}} |\Omega'| \quad (4.5)$$

where $r = \frac{1}{2} \dim \ker(I - P)$.

Let M be a symplectic manifold, Φ a symplectic diffeomorphism and Z a clean fixed point set of Φ . By Lemma 4.3 the tangent space to Z at each $z \in Z$ possesses an intrinsic positive density, or what is the same thing, Z possesses an intrinsic smooth positive measure, which we'll denote by $d\mu_Z$. In particular let Z be a submanifold of S^*X consisting of periodic H_q solution curves of period T . If Z is a clean fixed point set for $\Phi^T: S^*X \rightarrow S^*X$, then $Z' = \{(x, \lambda\xi), \lambda \in \mathbb{R}^+, (x, \xi) \in Z\}$ is a clean fixed point set for $\Phi^T: T^*X \searrow 0 \rightarrow T^*X \searrow 0$; so $d\mu_{Z'}$ is defined. Dividing by $|dq|$ we get an intrinsic measure $d\mu_Z$ on Z . We now state a result relating the integral of this measure over Z to the singular behavior of $\hat{\sigma}(t)$ at $t = T$.

Theorem 4.5. *Assume that the set of periodic H_q solution curves of period T is a union of connected submanifolds $Z_1, Z_2, \dots, Z_r$ in S^*X , each Z_j being a clean fixed point set for Φ^T of dimension d_j . Then there is an interval around T in which no*

other periods occur, and on such an interval we have $\hat{\sigma}(t) = \sum_{j=1}^r \beta_j(t - T)$ where

$$\beta_j(t) = \int_{-\infty}^{\infty} \alpha_j(s) e^{-ist} ds, \quad (4.6)$$

with

$$\alpha_j(s) \sim \left(\frac{s}{2\pi i} \right)^{(d_j-1)/2} i^{-\sigma_j} \sum_{k=0}^{\infty} \alpha_{j,k} s^{-k} \text{ as } s \rightarrow \infty. \quad (4.7)$$

Here we interpret $(1/i)^{(d_j-1)/2} = e^{-\pi i(d_j-1)/4}$ if d_j is even, and σ_j is an integer on which we will comment further at the end of Section 6. Finally

$$\alpha_{j,0} = \frac{1}{2\pi} \int e^{-iT \cdot \gamma} d\mu_{Z_j}, \quad (4.8)$$

with $\gamma(x, \xi) = \text{average of sub } Q \text{ over the periodic } H_q \text{ solution curve passing through } (x, \xi) \in S^*X$. Of course $\alpha_j(s) \sim 0$ as $s \rightarrow -\infty$.

If $d_j=1$ then the cleanness condition reduces to the condition that $I - P_j$ is invertible. Here P_j denotes the differential of Φ^T at $(x, \xi) \in Z_j$ reduced modulo the eigenvector $H_q(x, \xi)$. P_j will be called the (linear) Poincaré map. By Lemma 4.4, $d\mu_j = |I - P_j|^{-\frac{1}{2}} |dt|$, and integrating over Z_j gives

$$\alpha_{j,0} = \frac{T_0}{2\pi} e^{-iT \cdot \gamma} \cdot |I - P_j|^{-\frac{1}{2}},$$

where T_0 denotes the primitive period of the H_q -solution curve having Z_j as its orbit.

In particular, if $Q = (\lambda + c)^{\frac{1}{2}}$, then, for generic Riemannian metrics, the Poincaré maps of periodic H_q solution curves have no eigenvalues equal to 1 and for each period $T \neq 0$ there is not more than one antipodal pair of orbits of periodic H_q solution curves with that period. (See Klingenberg and Takens [25].) This situation has also been considered by Colin de Verdière [7]. The contributions from the antipodal orbits are equal; and, recalling that $\text{sub } Q = 0$ and $\int_0^\infty e^{-ist} ds = \frac{i}{t} + \frac{\delta}{4\pi}$, one gets from (4.6)–(4.8) the residue formula

$$\lim_{t \rightarrow T} (t - T) \hat{\sigma}(t) = \frac{T_0}{\pi} i^{-\sigma_j} |I - P_j|^{-\frac{1}{2}}. \quad (4.9)$$

Note that the primitive period, T_0 , is equal to the length of the geodesic covered by the periodic H_q solution curves with period equal to T .

5. Clean Intersection Theory

The proof of Theorem 4.5 will require some facts concerning composition of Fourier integrals under hypotheses less restrictive than those considered in Hörmander [17]. First, however, we'll need some elementary facts about symplectic vector spaces.

Let V and W be symplectic vector spaces, let Γ be a Lagrangian subspace of $V \times W$, and let A be a Lagrangian subspace of W . Let $\Gamma \circ A$ be the set of vectors $v \in V$, such that there exists $(v, w) \in \Gamma$ with $w \in A$.

Lemma 5.1. $\Gamma \circ \Lambda$ is a Lagrangian subspace of W .

Proof. Let ρ and π be the projections of Γ on V and W respectively. Consider the diagram

$$\begin{array}{ccc} \Gamma & \xleftarrow{\quad} & F \\ \pi \downarrow & & \downarrow \\ W & \xleftarrow{i} & \Lambda \end{array}$$

F being the fiber product: $\{(a, b) \in \Gamma \times \Lambda, \pi(a) = i(b)\}$. Associated with this diagram is an exact sequence

$$0 \rightarrow F \rightarrow \Gamma \oplus \Lambda \xrightarrow{\tau} W \rightarrow \text{coker } \tau \rightarrow 0 \quad (5.1)$$

where $\tau(a, b) = \pi(a) - i(b)$. $\Gamma \circ \Lambda$ can be defined as the image of the composite map

$$F \rightarrow \Gamma \xrightarrow{\rho} V.$$

Denoting this composite map by α , we get an exact sequence

$$0 \rightarrow \ker \alpha \rightarrow F \xrightarrow{\alpha} \Gamma \circ \Lambda \rightarrow 0. \quad (5.2)$$

We will now show that $\ker \alpha$ and $\text{coker } \tau$ are dually paired by the symplectic structure on W . Note first of all that $\ker \alpha$ consists of all pairs (a, w) in the fiber product for which $\rho(a) = 0$. We can write $a \in V \times W$ as a pair (v', w') . To say that (a, w) is in the fiber product says that $w' = w$, and to say $\rho(a) = 0$ says that $v' = 0$; so $\ker \alpha$ can be identified with the set of $w \in W$ such that

i) $w \in \Lambda$

and

ii) $(0, w) \in \Gamma$.

Suppose now that u is in the image of τ , i.e. $u = w_1 + \pi(v_2, w_2)$ with $(v_2, w_2) \in \Gamma$. Then $\Omega_W(w, w_1) = 0$ by i) and $\Omega_W(w, w_2) = 0$ by ii) so $\Omega_W(w, u) = 0$. Since Γ and Λ are maximally isotropic, this argument works backward to show that $(\ker \alpha)^\perp = (\text{Im } \tau)$ in W . It is easy now to see that the dimension of $\Gamma \circ \Lambda$ is half the dimension of V using the exact sequences (5.1) and (5.2). It is clear that $\Gamma \circ \Lambda$ is isotropic; thus it is Lagrangian, proving Lemma 5.1.

Lemma 5.2. Let $\alpha: F \rightarrow \Gamma \circ \Lambda$ be the mapping defined by (5.2). Then there is a canonical mapping of half-densities

$$|\Lambda|^{\frac{1}{2}} \otimes |\Gamma|^{\frac{1}{2}} \xrightarrow{\cong} |\Gamma \circ \Lambda|^{\frac{1}{2}} \otimes |\ker \alpha|.$$

Proof. From (5.1) we get an identification

$$|F|^{-\frac{1}{2}} \otimes |\Gamma|^{\frac{1}{2}} \otimes |\Lambda|^{\frac{1}{2}} \otimes |W|^{-\frac{1}{2}} \otimes |\text{coker } \tau|^{\frac{1}{2}} \cong 1$$

and

$$|F|^{\frac{1}{2}} \otimes |W|^{\frac{1}{2}} \otimes |\text{coker } \tau|^{-\frac{1}{2}} \cong |\Gamma|^{\frac{1}{2}} \otimes |\Lambda|^{\frac{1}{2}}.$$

From (5.2) we get

$$|F|^{\frac{1}{2}} \cong |\Gamma \circ \Lambda|^{\frac{1}{2}} \otimes |\ker \alpha|^{\frac{1}{2}}. \quad (5.3)(i)$$

The symplectic structure on W gives us a trivialization

$$|W|^{\frac{1}{2}} \cong 1 \quad (5.3)(ii)$$

and finally the dual pairing of $\ker \alpha$ and $\operatorname{coker} \tau$ via the symplectic structure of V gives us a mapping

$$|\operatorname{coker} \tau|^{-\frac{1}{2}} \cong |\ker \alpha|^{\frac{1}{2}}. \quad (5.3)(iii)$$

Putting this all together we get the assertion of Lemma 5.2. Q.E.D.

Remark. Let Γ' be the set of vectors $\{(v, -w), (v, w) \in \Gamma\}$. This is a Lagrangian subspace of $V \times W'$ where $W' = W$ as a vector space but has $-\Omega_W$ as its symplectic two-form. The Lagrangian subspaces of W and W' are the same; so Lemmas 5.1 and 5.2 are true for Γ' as well as Γ .

Given manifolds X, Y, Z and maps $f: X \rightarrow Z$ and $g: Y \rightarrow Z$, f and g are said to intersect *cleanly* (see [5]), if the fiber product

$$F = \{(x, y) \in X \times Y, f(x) = g(y)\}$$

represented by the diagram

$$\begin{array}{ccc} X & \xleftarrow{\quad} & F \\ f \downarrow & & \downarrow \\ Z & \xleftarrow{\quad g \quad} & Y \end{array} \quad (5.4)$$

is a submanifold of $X \times Y$, and in addition for each $p \in F$, $p = (x, y)$,

$$\begin{array}{ccc} T_x X & \xleftarrow{\quad} & T_p F \\ df_x \downarrow & & \downarrow \\ T_z Z & \xleftarrow{\quad dg_y \quad} & T_y Y \end{array}$$

is a fiber product diagram, i.e. $T_p F$ is the fiber product of $T_x X$ and $T_y Y$. (For example, if f and g intersect transversally, they intersect cleanly.) We associate to the diagram (5.4) a non-negative integer, e , called its *excess*:

$$e = \dim F + \dim Z - (\dim X + \dim Y).$$

Note that $e = 0$ if and only if the clean diagram (5.4) is transversal.

Now let X and Y be compact manifolds and Γ and Λ closed homogeneous Lagrangian submanifolds of $T^*(X \times Y) \setminus 0$ and $T^*Y \setminus 0$ respectively. Let $\Gamma' = \{(x, \xi, y, \eta), (x, \xi, y, -\eta) \in \Gamma\}$ and let $\Gamma' \circ \Lambda = \{(x, \xi), \exists (x, \xi, y, \eta) \in \Gamma', (y, \eta) \in \Lambda\}$. Let ρ and π be the projections of Γ' on T^*X and T^*Y . Assume neither $\Gamma' \circ \Lambda$ nor $\pi(\Gamma')$ contain zero vectors.

Lemma 5.3. *If the fiber product diagram*

$$\begin{array}{ccc} \Gamma' & \xleftarrow{\quad} & F \\ \pi \downarrow & & \downarrow \\ T^*Y & \xleftarrow{\quad} & \Lambda \end{array} \quad (5.5)$$

is clean, then $\Gamma' \circ A$ is an (immersed) Lagrangian submanifold of $T^*X \setminus 0$, and the composite map, α ,

$$F \rightarrow \Gamma' \xrightarrow{\rho} \Gamma' \circ A$$

is a fiber mapping with compact fiber.

Proof. The first part of the Lemma is just a rephrasing of Lemma 5.1. The fibers of $\alpha: F \rightarrow \Gamma' \circ A$ must be compact, for otherwise in view of the homogeneity of Γ' and A , $\Gamma' \circ A$ would have to contain zero vectors. Q.E.D.

Given half-densities on Γ and A , then by Lemma 5.2 we get an object on F which is a half-density in the horizontal direction times a density in the fiber direction. Integrating this over the fibers, we get a half-density on $\Gamma' \circ A$ which we will call the *composite* of the given densities on Γ and A . It is easy to check that if σ and τ are homogeneous densities, then so is their composite, $\sigma \circ \tau$, and

$$\text{degree } \sigma \circ \tau = \text{degree } \sigma + \text{degree } \tau - \frac{(\dim Y - e)}{2}, \quad (5.6)$$

e being the excess in the diagram (5.5). (The excess comes in because of the pairing (5.3)(iii), which is homogeneous of degree $e/2$.)

We now state a “clean intersection” theorem on composition of Fourier integrals. For basic facts concerning oscillatory integrals and the spaces, $I^k(Y, A)$, see Hörmander [17].

Theorem 5.4. *Let k be a generalized half-density on $X \times Y$ and K the operator associated with it. (K maps compactly supported half-densities on Y to generalized half-densities on X .) Let Γ and A be as above. If k belongs to $I^m(X \times Y, \Gamma)$; then K maps $I^s(Y, A)$ into*

$$I^{m+s+\frac{e}{2}}(X, \Gamma' \circ A) \quad \text{and} \quad \sigma(K\mu) = (2\pi i)^{-\frac{e}{2}} \sigma(k) \circ \sigma(\mu)$$

modulo Maslov factors.

We will briefly describe how to handle the Maslov factors. Let L_Γ , L_A and $L_{\Gamma' \circ A}$ be the Maslov bundles of Γ , A and $\Gamma' \circ A$. Let $L_{\Gamma'}$ be the pull back of L_Γ to Γ' via $(x, \xi, y, \eta) \rightarrow (x, \xi, y, -\eta)$. Let $i: F \rightarrow \Gamma' \times A$ be the inclusion map and $\alpha: F \rightarrow \Gamma' \circ A$ defined above. We claim

$$i^*(L_{\Gamma'} \boxtimes L_A) \cong \alpha^* L_{\Gamma' \circ A}. \quad (5.7)$$

Granting (5.7) it is clear how to define $\sigma(k) \circ \sigma(\mu)$ when $\sigma(k)$ and $\sigma(\mu)$ are half-densities tensored with sections of the Maslov bundle. In fact the product of $\sigma(k)$ with $\sigma(\mu)$ is now an object on F which is a density vertically and a half-density times a section of $\alpha^* L_{\Gamma' \circ A}$ horizontally. Integrating over the fiber we get an object on $\Gamma' \circ A$ which is a half-density tensored with a section of $L_{\Gamma' \circ A}$.

To prove (5.7) let $\varphi_1 = \varphi_1(x, y, \xi)$ be a phase function for Γ and $\psi_1 = \psi_1(y, \eta)$ a phase function for A . Then $\varphi_1 + \psi_1$ is a phase function for $\Gamma' \circ A$ (with $\omega = (y, \xi, \eta)$ as phase variables).² If $\varphi_2 = \varphi_2(x, y, \xi')$ and $\psi_2 = \psi_2(y, \xi')$ are another pair of such

² $\psi + \varphi$ can be degenerate; however it is a “clean” phase function in the sense of § 7. Such phase functions can be used to give a Čech-theoretic definition of the Maslov class, just as Hörmander does in § 3.2 of [17].

phase functions, then on the overlaps of their domains of definition

$$\begin{aligned} & \operatorname{sgn}(\varphi_1 + \psi_1)''_{\omega\omega} - \operatorname{sgn}(\varphi_2 + \psi_2)''_{\omega'\omega'} \\ &= \operatorname{sgn}(\varphi_1)''_{\xi\xi} - \operatorname{sgn}(\varphi_2)''_{\xi'\xi'} + \operatorname{sgn}(\psi_1)''_{\eta\eta} - \operatorname{sgn}(\psi_2)''_{\eta'\eta'}. \end{aligned} \quad (5.8)$$

(*Proof.* (5.8) is obvious when φ_1, φ_2 and ψ_1, ψ_2 are equivalent. It is also true if we modify such equivalent pairs by adding to φ_2 and ψ_2 quadratic forms in new phase variables. By Hörmander [17] §3.1 this covers all possible cases.) (5.8) implies that the transition functions of $L_{T^*\circ A}$ are the products of the transition functions of L_T and L_A . (5.7) follows.

6. Proof of Theorem 4.5

This will be by applying Theorem 5.4 to the distribution $U(t, x, y)$ of Theorem 1.1. For this we need to know the symbol of U . We know already from Theorem 1.1 that $U \in I^{-\frac{1}{2}}(\mathbb{R} \times X \times X; C')$, C being the set (1.15). Its symbol is a section of the half-density bundle of C' times a section of the Maslov bundle of C' . To see what it is, first note that both these bundles have intrinsic trivializations. In fact, for the half density bundle, the projection $\pi: C \rightarrow (T^*X \setminus 0) \times \mathbb{R}$, $\pi(t, \tau, x, \xi, y, \eta) = (t, x, \xi)$ is a diffeomorphism; so

$$\sigma = \pi^*(|dt|^{\frac{1}{2}} \otimes |dx \wedge d\xi|^{\frac{1}{2}}) \quad (6.1)$$

is a nowhere vanishing half-density.

Consider next $L_{C'}$. $L_{C'}$ is a locally constant line bundle (i.e. its transition functions are constant); so we are interested in finding a trivialization given by a *constant* section. Such a section, if it exists, is unique up to scalar multiples. To see that such a section exists note that the subset $t=0$ of C' is identical with $N^*\Delta = \{(x, \xi, x, -\xi), (x, \xi) \in T^*X \setminus 0\}$. Moreover $L_{C'}$ restricted to $N^*\Delta$ is its Maslov bundle. Since $N^*\Delta$ is a normal bundle, $L_{N^*\Delta}$ possesses a canonical constant section, s , by Hörmander [17] §3.3. Now extend s to a global section of $L_{C'}$ by requiring it to be constant along each bicharacteristic:

$$(t, \tau, x, \xi, y, -\eta), (x, \xi) = \Phi^t(y, \eta) \quad (6.2)$$

with $-\infty < t < \infty$. It is clear this can be done in one and just one way.

Now let σ be the section of $\Omega_{\frac{1}{2}} \otimes L_C$ obtained by tensoring (6.1) with the constant section of $L_{C'}$ just described.

Lemma 6.1.

$$\sigma(U) = \sigma \cdot \exp \left(-i \int_0^t \operatorname{sub} Q(\Phi^s(y, \eta)) ds \right)$$

at the point $((t, \tau), (x, \xi), (y, -\eta)) \in C'$.

Proof. At $t=0$ the left and right hand side are equal since both are equal to the symbol of the identity map. Moreover σ is invariant under the Hamilton flow so the right hand side satisfies the transport equation for $\sigma(U)$ induced by the equation $(i^{-1} \partial/\partial t + Q) \cdot U = 0$. (See [9], Theorem 5.3.1.) This proves the equality for all t .

Let $\Delta: \mathbb{R} \times X \rightarrow \mathbb{R} \times X \times X$ be the diagonal map. Given a half density u on $\mathbb{R} \times X \times X$ we can pull it back to the diagonal and multiply the two half density factors in X to get an object $\Delta^* u$ which is a density in X times a half-density in $\mathbb{R}$ (at each point of $X \times \mathbb{R}$). This object can be integrated over X to get a half-density on $\mathbb{R}$ which we will denote by $\pi_* \Delta^* u$ ($\pi: \mathbb{R} \times X \rightarrow \mathbb{R}$ being the projection map). Since $\pi_* \Delta^*$ is an operator from half densities on $\mathbb{R} \times X \times X$ to half densities on $\mathbb{R}$ its Schwartz kernel, $k_{\pi_* \Delta^*}$, is a half density on $\mathbb{R} \times X \times X \times \mathbb{R}$. In the following lemma, identify $\mathbb{R} \times X \times X \times \mathbb{R}$ with $(\mathbb{R} \times X) \times (\mathbb{R} \times X)$ via $(t, x, y, s) \rightarrow (s, x, t, y)$.

Lemma 6.2. $k_{\pi_* \Delta^*} = k_{\text{id}}$, k_{id} being the Schwartz kernel of the identity map.

Proof. Both $k_{\pi_* \Delta^*}$ and k_{id} can be viewed as multi-linear functionals on

$$C^\infty(\mathbb{R}, \Omega_{\frac{1}{2}}) \times C^\infty(X, \Omega_{\frac{1}{2}}) \times C^\infty(X, \Omega_{\frac{1}{2}}) \times C^\infty(\mathbb{R}, \Omega_{\frac{1}{2}}).$$

It is clear that they are identical with the multi-linear functional given by the pre-Hilbert structures on $C^\infty(X, \Omega_{\frac{1}{2}})$ and $C^\infty(\mathbb{R}, \Omega_{\frac{1}{2}})$. Q.E.D.

Let Γ be the conormal bundle to the diagonal in $\mathbb{R} \times X \times \mathbb{R} \times X$. We will show, as we did above for C , that both the half-density bundle and the Maslov bundle of Γ are trivial. To see that the first of these assertions is true, note that the projection map $\pi: \Gamma \rightarrow T^*(\mathbb{R} \times X) \setminus 0$, $\pi(t, \tau, x, \xi, x, -\xi, t, -\tau) = (t, \tau, x, \xi)$ is bijective and set

$$\sigma' = \pi^* |dt \wedge d\tau \wedge dx \wedge d\xi|^{\frac{1}{2}} \quad (6.3)$$

a nowhere vanishing half density on Γ .

The Maslov bundle of Γ is trivial since Γ is a conormal bundle; however, it will be helpful to see explicitly why. Over any coordinate patch, Γ is describable by the phase function

$$\Psi = (x - y) \cdot \xi + (t - s)\tau. \quad (6.4)$$

It is clear that any two such Ψ 's are equivalent; so the set of all such Ψ 's determines a unique non-zero constant section of L_Γ .

Lemma 6.3. $k_{\pi_* \Delta^*} \in I^0(\mathbb{R} \times X \times X \times \mathbb{R}, \Gamma)$ and $\sigma(k_{\pi_* \Delta^*}) = \sigma'$.

Proof. Both these statements are well-known for k_{id} . See, for example, Hörmander [17], Section 4.

Going back to (1.10) of Section 1 we have

$$U(x, y, t) = \sum e^{iu_j t} e_j(x) \bar{e}_j(y)$$

where $e_j \in C^\infty(X, \Omega_{\frac{1}{2}})$ is the j -th eigenfunction and u_j the j -th eigenvalue of Q . Setting $x = y$ and integrating over X we get

$$\sum e^{-iu_j t} \langle e_j, e_j \rangle_{L^2} = \sum e^{-iu_j t};$$

so this proves

Lemma 6.4. $\sum e^{-iu_j t} = \pi_* \Delta^* U$.

We are now in position to apply the clean intersection theory of Section 5. To do so we need to know that the fiber product diagram

$$\begin{array}{ccc} \Gamma' & \longleftarrow & \Gamma \\ \downarrow & & \downarrow \\ T^*(\mathbb{R} \times X \times X) & \longleftarrow & C \end{array} \quad (6.5)$$

is clean. The map of Γ' into $T^*(\mathbb{R} \times X \times X)$ is an embedding, and the image consists of the points $\{(t, \tau, x, \xi, x, -\xi)\}$. To apply Theorem 5.4 we need to know this intersects C cleanly. By (1.15) the intersection is the set of all (T, x, ξ) for which $\Phi^T(x, \xi) = (x, \xi)$, i.e. the fixed point set of Φ^T ; and by definition, Γ' intersects C' cleanly if and only if this fixed point set is clean. Suppose the fixed point set is a d -dimensional submanifold of S^*X . d is the excess in diagram (6.5); so combining Theorem 5.4, Lemma 6.3 and Lemma 6.1 we get

$$\pi_* \Delta^* U \in I^{\frac{d}{2} - \frac{1}{4}}(\Lambda_T),$$

where

$$\Lambda_T = \{(T, \tau), \tau \in \mathbb{R}^-\}.$$

By definition, $I^{\frac{d}{2} - \frac{1}{4}}(\Lambda_T)$ consists of scalar multiplies of the distribution

$$\frac{1}{2\pi} \int_0^\infty s^{\frac{d-1}{2}} e^{-is(t-T)} ds \quad (6.6)$$

plus similar distributions of lower order. The coefficient of (6.6) is the symbol of $\pi_* \Delta^* U$. By Theorem 5.4 this is equal to $\sigma(k_{\pi_* \Delta^*}) \circ \sigma(u)$. The half-density part of this is easily computed to be

$$s^{\frac{d-1}{2}} \sqrt{ds} \int e^{i\gamma T} d\mu_Z \quad (6.7)$$

(just observe that (4.2) is a special case of Lemma 5.2 with Γ equal to the diagonal and Λ equal to graph P).

To compute the Maslov contribution to (6.9), represent $U(t, x, y)$ locally by oscillatory integrals (1.17) with phase functions of the form

$$\varphi(t, x, y, \eta) = \chi(t, x, \eta) - \langle y, \eta \rangle. \quad (6.8)$$

This can be done by taking $\chi(t_0, x, \eta)$ as the generating function for the canonical transformation Φ^{t_0} , that is

$$\text{graph}(\Phi^{t_0})^{-1} = \left\{ \left(\begin{pmatrix} x \\ d_x \chi(t_0, x, \eta) \end{pmatrix}, \begin{pmatrix} d_\eta \chi(t_0, x, \eta) \\ \eta \end{pmatrix} \right); \quad x, \eta \in \mathbb{R}^n \right\}, \quad (6.9)$$

and then taking χ for nearly t as the solution of

$$d_t \chi(t, x, \eta) + q(x, d_x \chi(t, x, \eta)) = 0. \quad (6.10)$$

If we do this near a period $t_0 = T$ then the leading term in (1.25) is equal to a positive constant times $e^{s\pi i/4} a$, where s is equal to the signature of the matrix

$$d_{(x,t,\eta)}^2 \varphi(t, x, \eta) = \begin{pmatrix} d_t^2 \chi & d_x d_t \chi & d_\eta d_t \chi \\ d_t d_x \chi & d_x^2 \chi & d_\eta d_x \chi - I \\ d_t d_\eta \chi & d_x d_\eta \chi - I & d_\eta^2 \chi \end{pmatrix}. \quad (6.11)$$

One can choose the local coordinates such that at the periodic point one has $\partial/\partial x_1 = H_q$ and $\partial/\partial \xi_1$ is tangent to the cone axis. Differentiating (6.10) and the equation $q(x, d_x \chi(t, x, \eta)) = q(d_\eta \chi(t, x, \eta), \eta)$ it follows that (6.11) has the matrix

$$\begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

on the space spanned by $\partial/\partial t$, $\partial/\partial x_1$, $\partial/\partial \xi_1$, and that the vectors $\partial/\partial x_j$, $\partial/\partial \xi_j$ for $j > 1$ are orthogonal to this space with respect to (6.11). So s is equal to the signature of the matrix

$$Q = \begin{pmatrix} d_x^2 \chi & d_\eta d_x \chi - I \\ d_x d_\eta \chi - I & d_\eta^2 \chi \end{pmatrix} \quad (6.12)$$

restricted to the space spanned by $\partial/\partial x_j$, $\partial/\partial \xi_j$ with $j > 1$.

Next we have to investigate the Maslov factors picked up by the amplitude a . Let $\varphi_j = \chi_j - \langle y, \eta \rangle$, $j = 1, \dots, J$ be a sequence of phase functions of the form (6.8) related to a chain of neighborhoods in C' along the $H_{(\tau+q)}$ solution curve; denote the corresponding amplitudes by a_j . Let t_j , $j = 1, \dots, J-1$, be times at which the curve is in the intersection of the j -th and $(j+1)$ -th neighborhood. Using Hörmander [17, p. 148] we have at time t_j :

$$a_{j+1} = i^{\frac{1}{2}(\text{sgn } d_\eta^2 \varphi_j - \text{sgn } d_\eta^2 \varphi_{j+1})} a_j \quad (6.13)$$

modulo positive factors, so the amplitude picks up the factor i^k with

$$k = \sum_{j=1}^{J-1} \frac{1}{2} (\text{sgn } d_\eta^2 \varphi_j - \text{sgn } d_\eta^2 \varphi_{j+1}). \quad (6.14)$$

Now $d_\eta^2 \varphi_j = d_\eta^2 \chi_j$, and according to (6.9)

$$\left\{ \begin{pmatrix} d_\eta^2 \chi_j & \delta \eta \\ \delta \eta & \end{pmatrix}; \delta \eta \in \mathbb{R}^n \right\} = (d\Phi^{t_j})^{-1}(V)$$

where V denotes the “vertical space” $\left\{ \begin{pmatrix} \delta x \\ \delta \xi \end{pmatrix}; \delta x = 0 \right\}$, that is, V is the tangent space to the fiber in T^*X . Denote by H_j the “horizontal space” $\left\{ \begin{pmatrix} \delta x \\ \delta \xi \end{pmatrix}; \delta \xi = 0 \right\}$ for the local coordinatization of X used in the j -th local representation of $U(t, x, y)$. Then $(d\Phi^{t_j})^{-1}(V)$ is the graph of the linear mapping $d_\eta^2 \chi_j: V \rightarrow H_j$, and it follows that

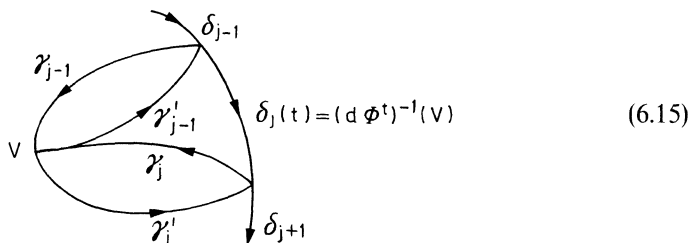
$$d_\eta^2 \chi_j = -Q(V, H_j; (d\Phi^{t_j})^{-1}(V))$$

in the notation of [12, Section 2]. Using [12, (2.10') and (2.13)] it follows that

$$\begin{aligned} & \frac{1}{2}(\operatorname{sgn} d_{\eta}^2 \varphi_j - \operatorname{sgn} d_{\eta}^2 \varphi_{j+1}) \\ &= \frac{1}{2} \{ \operatorname{sgn} Q(V, H_j; (d\Phi^t)^{-1}(V)) - \operatorname{sgn} Q(V, H_{j+1}; (d\Phi^t)^{-1}(V)) \} \\ &= -s((d\Phi^t)^{-1}(V), V; H_j, H_{j+1}) = s(H_j, H_{j+1}; (d\Phi^t)^{-1}(V), V). \end{aligned}$$

This integer in turn is equal to the Maslov-Arnol'd index of the loop consisting of an arc γ_j of Lagrange spaces going from $(d\Phi^t)^{-1}(V)$ to V transversal to H_j followed by an arc γ'_j of Lagrange spaces going from V to $(d\Phi^t)^{-1}(V)$ transversal to H_{j+1} .

Denote by δ_j the curve $(d\Phi^t)^{-1}(V)$, t running from t_{j-1} to t_j , δ_1 the same one for t running from 0 to t_1 and δ_J this curve for t running from t_{J-1} to the period T . By construction $\delta_j(t)$ is transversal to H_j for $t \in [t_{j-1}, t_j]$ and we can arrange that δ_1, δ_J both are transversal to $H_1 = H_J$. Write δ for the curve $(d\Phi^t)^{-1}(V)$, t running from 0 to T and ε for the curve $\delta_J \circ \gamma'_{J-1} \circ \gamma_{J-1} \circ \delta_{J-1} \circ \dots \circ \delta_2 \circ \gamma'_1 \circ \gamma_1 \circ \delta_1$.



Because the loop $\gamma_J \circ \delta_J \circ \gamma'_{J-1}$ is contained in the contractible set of Lagrange spaces transversal to H_j its intersection number with any Lagrange space H vanishes. Because $\delta_J, \gamma'_{J-1}, \gamma_1, \delta_1$ are transversal to H_1 it follows that the intersection number $[\varepsilon; H_1]$ of the curve ε with H_1 vanishes too. On the other hand $[\varepsilon; H_1] = \sum_{j=1}^{J-1} [\gamma'_1 \circ \gamma_J; H_1] + [\delta; H_1] = k + [\delta; H_1]$. We conclude that the symbol of $\pi_* \mathcal{A}^* U$ is equal to (6.7) multiplied by

$$e^{-\frac{\pi i}{2}} \{ [\delta; H_1] - \frac{1}{2} \operatorname{sgn} Q \} = i^{-\sigma} e^{-\pi i(d-1)/4} \quad (6.16)$$

if we write

$$\begin{aligned} \sigma &= [\delta; H_1] + \operatorname{ind} Q - (n-1) \\ &= [\delta; H_1] + \frac{1}{2}(\operatorname{rank} Q - \operatorname{sgn} Q) - (n-1) \\ &= [\delta; H_1] - \frac{1}{2} \operatorname{sgn} Q - \frac{1}{2} \operatorname{nullity} Q, \end{aligned}$$

and note that $\operatorname{nullity} Q = d-1$.

This completes the proof of Theorem 4.5. The integer σ appearing in the exponent of i can be identified with the Morse index for periodic geodesics in the following case. Assume that $d_{\xi}^2 q$ is positive definite on $\operatorname{Ker} d_{\xi} q$ at points (x, ξ) on the orbit of the periodic H_q solution curves. This means that the sphere $q=1$ in the fiber is strictly convex (as a boundary of the ball $q \leq 1$) at these points (x, ξ) .

Taking $p = q^m$ for some $m > 1$ it follows that $d_\xi^2 p$ is positive definite and $\frac{1}{m} H_p$ is related via the Legendre transformation to the Euler-Lagrange flow for a variational problem with finite index (see [12]).

Denoting the $\frac{1}{m} H_p$ flow by $\tilde{\Phi}^t$, the matrix of $(d\tilde{\Phi}^t)^{-1}$ on the space spanned by $\frac{1}{m} H_p$ and the direction of the cone axis is of the form $\begin{pmatrix} 1 & ct \\ 0 & 1 \end{pmatrix}$ with $c > 0$. (Note that $\tilde{\Phi}^t = \Phi^t$ on S^*X .) If $\tilde{\Phi}^T$ is represented as in (6.9) with a generating function $\tilde{\chi}$, then $\text{ind } \tilde{Q} = \text{ind } Q + 1$ if we write

$$\tilde{Q} = \begin{pmatrix} d_x^2 \tilde{\chi} & d_\eta d_x \tilde{\chi} - I \\ d_x d_\eta \tilde{\chi} - I & d_\eta^2 \tilde{\chi} \end{pmatrix} \quad \text{on } \mathbb{R}^{2n}$$

and $T > 0$. So $\sigma = [\delta: H_1] + \text{ind } \tilde{Q} - n$ and according to [12, Proposition 4.6 and formula (4.5)] this number is equal to the Morse index for the variational problem with periodic boundary conditions. This identification was suggested to us by the expansion of Colin de Verdière [7] who used the trace of the heat operator rather than the wave operator.

7. The Proof of Theorem 5.4

We begin by considering oscillatory integrals of the type considered by Hörmander in [17], i.e. integrals of the form

$$\mu(x) = \int a(x, \theta) e^{i\varphi(x, \theta)} d\theta \quad (7.1)$$

where $a(x, \theta)$ and $\varphi(x, \theta)$ are smooth functions on $X \times \mathbb{R}^N$ with the properties

(1) $\varphi(x, \theta)$ is homogeneous of degree one in θ and $d\varphi \neq 0$.

(2) $a(x, \theta) = 0$ near $X \times \{0\}$ and is homogeneous of degree $d + (n - 2N)/4$ for θ large.

Finally there is a third condition which Hörmander imposes on φ :

(3) If $\frac{\partial \varphi}{\partial \theta}(x, \theta) = 0$ then at (x, θ) the differentials $d\left(\frac{\partial \varphi}{\partial \theta_1}\right), \dots, d\left(\frac{\partial \varphi}{\partial \theta_N}\right)$ are linearly independent.

Let C_φ be the set of points where $\frac{\partial \varphi}{\partial \theta} = 0$. Then condition (3) implies that C_φ is an n -dimensional submanifold of $X \times \mathbb{R}^N$ and that the map

$$C_\varphi \rightarrow T^*X, \quad (x, \theta) \rightarrow \frac{\partial \varphi}{\partial x} \quad (7.2)$$

immerses C_φ as a Lagrangian submanifold, Λ , of T^*X . $I^d(X, \Lambda)$ is defined to be the space of all distributions which have a local representation of the form (7.1). The symbol of (7.1) is defined as follows. Let d_C be the density on C_φ obtained by pulling back the delta function on $\mathbb{R}^N$ via $X \times \mathbb{R}^N \rightarrow \mathbb{R}^N$, $(x, \theta) \rightarrow \varphi'_\theta$. The half density part of the symbol is the image of

$$(2\pi)^{-n} (2\pi i)^{-\frac{N}{2}} a \sqrt{d_C}$$

with respect to (7.2), and the Maslov part is the section of L_Λ determined by φ .

For our purposes we need to consider distributions of the type (7.1) satisfying (1) and (2), but with (3) replaced by a weaker “cleanness” condition, to wit:

(3)' C_φ is a submanifold of $X \times \mathbb{R}^N$ and at each point of C_φ the tangent space is the space of vectors annihilated by $d\left(\frac{\partial\varphi}{\partial\theta_1}\right), \dots, d\left(\frac{\partial\varphi}{\partial\theta_N}\right)$.

Suppose that the dimension of the space spanned by these differentials is $N - e$.

Lemma 7.1. *The map $C_\varphi \rightarrow T^*X$ defined by (7.2) has as its image an immersed Lagrangian manifold $A \subset T^*X$ and the map $C_\varphi \rightarrow A$ is a fiber mapping of fiber dimension e . Moreover the distribution $\mu(x)$ defined by (7.1) is in $I^{d+\frac{e}{2}}(X, A)$.*

Proof. To see that the image of C_φ is a Lagrangian manifold, we apply Lemma 5.3 of Section 5 with Λ replaced by $\tilde{\Lambda} = \text{graph } d\varphi$ and Γ replaced by the normal bundle to the graph of $\pi: X \times \mathbb{R}^N \rightarrow X$, which we will denote by H . It is simple to check that the fiber product diagram (5.5) of Section 5 is clean if and only if (3)' holds. To prove the last assertion we write (7.1) in polar coordinate form. Set $\theta = s\omega$ with $\omega \in S^{N-1}$. Then

$$\mu(x) = \int s^k a(x, \omega) e^{is\varphi(x, \omega)} d\omega ds \quad \text{where } k = d - 1 + (n + 2N)/4. \quad (7.3)$$

We can assume that $a(x, \omega)$ has its support in a coordinate patch (on S^{N-1}), hence that when the integrand in (7.3) is non-zero, ω is in a compact subset of S^{N-1} . The “polar” critical set C'_φ in $X \times \mathbb{R}^{N-1}$ is defined by the equations

$$\varphi = 0, \quad \frac{\partial\varphi}{\partial\omega_1} = 0, \quad \frac{\partial\varphi}{\partial\omega_2} = 0, \dots, \quad \frac{\partial\varphi}{\partial\omega_{N-1}} = 0. \quad (7.4)$$

By (3)' the differentials of these functions are the defining equations for the normal space at each point of C'_φ . By a change of coordinates we can assume that the first $N - e$ of these differentials are linearly independent, and that

$$d\varphi, d\left(\frac{\partial\varphi}{\partial\omega_1}\right), \dots, d\left(\frac{\partial\varphi}{\partial\omega_{l-1}}\right), d\omega_l, \dots, d\omega_{N-1}, \quad l = N - e,$$

are linearly independent. This implies that C'_φ is locally defined by the first l equations of (7.4) and that C'_φ intersects the surface $\omega_l = \text{const}$, $\omega_{l+1} = \text{const}$, ..., $\omega_{N-1} = \text{const}$ transversally. Let ω' denote the first $l - 1$ coordinates and ω'' the remaining e coordinates. Then for $\omega'' = c$, the function $\varphi(x, \omega', c)$ is a non-degenerate phase function on $X \times \mathbb{R}^{l-1}$ (in Hörmander's sense), and its critical set is just the intersection of C'_φ with $\omega'' = c$. So in particular its associated Lagrangian manifold in T^*X is A . Now write

$$\mu(x) = \left(\int s^k a(x, \omega', \omega'') e^{i\varphi(x, \omega', \omega'')} d\omega'' \right) d\omega'. \quad (7.5)$$

For fixed ω'' the inner integral is in $I^{d+\frac{e}{2}}(X, A)$; therefore, so is μ itself.

To compute the symbol of (7.5), identify the compact subset of $\mathbb{R}^{l-1}$ where ω' is defined with a subset of S^{l-1} by stereographic projection and set $\theta' \in \mathbb{R}^l$ equal to $s\omega'$. Let

$$a_{\omega''}(x, \theta') = s^k a(x, \omega', \omega'')$$

and

$$\varphi_{\omega''}(x, \theta') = s\varphi(x, \omega', \omega'').$$

Then the symbol of (7.5) is equal to

$$\int \sigma_{\omega''} d\omega'' \quad (7.6)$$

where $\sigma_{\omega''}$ is the image of the half density

$$(2\pi)^{-n} (2\pi i)^{-\left(\frac{N-e}{2}\right)} a_{\omega''} \sqrt{dC_{\omega''}}$$

with respect to

$$(d\varphi_{\omega''})_x: C_{\omega''} \rightarrow \Lambda.$$

This symbol can be described more intrinsically as follows. Let $\tilde{\Lambda}$ be the graph of $d\varphi$ in $T^*(X \times \mathbb{R}^N) \setminus 0$ and let H be the conormal bundle to the graph of $\pi: X \times \mathbb{R}^N \rightarrow X$ in $T^*(X \times (X \times \mathbb{R}^N)) \setminus 0$. Let α be the half density $\sqrt{dx d\theta}$ pulled back to $\tilde{\Lambda}$. Identify H with the subset, $\{(x, \xi, \theta, 0)\}$, of $T^*(X \times \mathbb{R}^N)$ and let $\beta = (\sqrt{dx \wedge d\xi}) \sqrt{d\theta}$. By assumption $\tilde{\Lambda}$ and H intersect cleanly, so by Lemma 5.3 the composite half density, $\beta \circ \alpha$, on $\Lambda = H \circ \tilde{\Lambda}$ is well defined.

Lemma 7.2. (7.6) is equal to $(2\pi)^{-N} (2\pi i)^{-\frac{N-e}{2}} \beta \circ \alpha$.

Proof. When $e=0$ this is just a paraphrase of the formula above. If $\varphi_{\omega''}$ is independent of ω'' the equivalence of the two terms reduces to the case $e=0$. The general case can be reduced to the case where $\varphi_{\omega''}$ is independent of ω'' by Hörmander [17], Theorem 3.1.6. (An inspection of Hörmander's proof shows that it applies to families of phase functions.) Q.E.D.

We now proceed to the proof of Theorem 5.4. With the notation of Theorem 5.4, let $\mu \in I_{\Lambda}^k$. Represent μ and k by oscillatory integrals:

$$k(x, y) = \int a(x, y, \theta) e^{i\varphi(x, y, \theta)} d\theta,$$

$$\mu(y) = \int b(y, \xi) e^{i\psi(y, \xi)} d\xi.$$

Then $K \cdot \mu$ is represented by the oscillatory integral

$$K\mu(x) = \int a(x, y, \theta) b(y, \xi) e^{i(\varphi(x, y, \theta) + \psi(y, \xi))} d\theta d\xi dy. \quad (7.7)$$

We claim that this is in $I^{m+k+\frac{e}{2}}(X, \Gamma' \circ \Lambda)$. The proof is practically identical with the proof of Theorem 4.2.2. in Hörmander [17], so we will just sketch its main outlines. The integral (7.7) is of course an oscillatory integral in the phase variables (ξ, θ, y) but neither its amplitude nor its phase function satisfy (1) and (2). In the first place the amplitude and phase function aren't homogeneous in $\omega = (\xi, \theta, y)$, for $|\omega|$ large and in the second place they are not even homogeneous in (ξ, θ) alone for $|\xi| + |\theta|$ large (i.e. there are problems when θ is close to zero or ξ is close to zero). The first difficulty can be handled by using polar coordinates like in the proof of Lemma 7.1. To handle the second difficulty note that the critical set of $\varphi + \psi$ with respect to ω , is the fiber product of C_{φ} and C_{ψ} :

$$\begin{array}{ccc} C_{\psi} & \longleftarrow & C_{\varphi + \psi} \\ \downarrow & & \downarrow \\ T^*Y & \longleftarrow & C_{\varphi} \end{array}$$

This set can't contain (ξ, θ) with either ξ or θ equal to zero since this would contradict the assumption that the projection of Γ' on T^*Y and $\Gamma' \circ A$ don't contain zero vectors. Therefore we can assume that there exists a constant $K > 0$ such that

$$K |\theta| < |\xi| < \frac{1}{K} |\theta|$$

on a neighborhood of the critical set of $\varphi + \psi$. Off this set, the integrand of (7.7) can be made as smooth as desired by integration by parts, and on this set $\varphi + \psi$ is a clean phase function in the sense of (3') as long as $|\xi| + |\theta|$ is large, so we can apply Lemma 7.1. (For further details we refer to Hörmander [17], Section 4.2.)

To compute the symbol of (7.7) we will need first to make some general observations concerning composition of canonical relations. Let $C_1 \subset T^*(X \times Y) \setminus 0$ and $C_2 \subset T^*(Y \times Z) \setminus 0$ be Lagrangian manifolds. Assume they are properly situated with respect to each other in the sense of Theorem 4.22 of Hörmander [17]. We will say they intersect *cleanly* if $C'_1 \times C'_2$ intersects cleanly the set of points $\{(x, \xi, y, \eta, y, \eta, z, \gamma)\}$ in $T^*X \times T^*Y \times T^*Y \times T^*Z$. This means that

$$\begin{array}{ccc} \Delta' & \longleftarrow & \text{fiber product} \\ \downarrow & & \downarrow \\ T^*(X \times Y \times Y \times Z) & \longleftarrow & C_1 \times C_2 \end{array}$$

is a clean diagram in the sense of (5.5) where Δ is the normal bundle to the diagonal in $T^*((X \times Y \times Y \times Z) \times X \times Z)$. By Lemma (5.3) $C'_1 \circ C'_2 = \Delta' \circ (C_1 \times C_2)$ is a Lagrangian submanifold of $T^*(X \times Z)$. It is just the set of all $(x, \xi, z, -\gamma)$ such that $\exists (y, \eta)$ with $(x, \xi, y, -\eta) \in C_1$ and $(y, \eta, z, -\gamma) \in C_2$. Given half densities σ_1 and σ_2 on C_1 and C_2 then by Lemma 5.3 we get a half density on $C_1 \circ C_2$ which we will call the composite half density and denote by $\sigma_1 \circ \sigma_2$. We leave it as an exercise to show that this composition is associative in the following sense

Lemma 7.3. *Let $C_1 \subset T^*(X \times Y) \setminus 0$, $C_2 \subset T^*(Y \times Z) \setminus 0$ and $A \subset T^*(Z) \setminus 0$. Assume C_1 , C_2 and A are all properly situated with respect to each other and all intersections are clean. Then*

$$(C'_1 \circ C'_2) \circ A = C'_1 \circ (C'_2 \circ A) \quad \text{and} \quad (\sigma_1 \circ \sigma_2) \circ \sigma = \sigma_1 \circ (\sigma_2 \circ \sigma)$$

for half densities σ_1, σ_2 and σ on C_1, C_2 and A .

Now let $\tilde{\Gamma}$ be the graph of $d\varphi$ in $T^*(X \times Y \times \mathbb{R}^N)$ and $\tilde{A}$ be the graph of $d\psi$ in $T^*(Y \times \mathbb{R}^M)$. Let H_{Γ} be the normal bundle to the graph of $\pi: X \times Y \times \mathbb{R}^N \rightarrow X \times Y$. (H_{Γ} is in $T^*((X \times Y \times \mathbb{R}^N) \times X \times Y)$) and let H_A be the graph of $\pi: Y \times \mathbb{R}^M \rightarrow Y$. Let $\tilde{\Delta}$, in $T^*((X \times Y \times Y) \times X)$, be the normal bundle to the diagonal. Let σ_{Δ} be the half density on $\tilde{\Delta}$ obtained by identifying $\tilde{\Delta}$ with $T^*(X \times Y)$. (Compare with Lemma 6.4.) Then $\Gamma' \circ A = \Delta' \circ (\Gamma \times A)$. Moreover if α and β are half densities on Γ and A , $\alpha \circ \beta = \sigma_{\Delta} \circ (\alpha \times \beta)$.

Consider the half densities $z(x, y, \theta) \sqrt{dx dy d\theta}$ on $\tilde{\Gamma}$ and $b(y, \xi) \sqrt{dy d\xi}$ on $\tilde{A}$. One way to get a half density on $\Gamma' \circ A$ is to take the product half density on $\tilde{\Gamma} \times \tilde{A}$, apply $H_{\Gamma} \times H_A$ to it to get a half density on $\Gamma \times A$ then apply $\tilde{\Delta}$ to get a half density

on $\Gamma' \circ \Lambda$. By Lemma 7.2 this is the half density associated with the integrand of (7.7). The other possibility is to apply H_r to the half density on $\tilde{\Gamma}$ (which by Lemma 7.2 is the symbol of k) apply H_Λ to the half-density on $\tilde{\Lambda}$ (which is the symbol of μ) then apply $\tilde{\Lambda}$ to the product, which is the same as $\sigma(k) \circ \sigma(\mu)$. The assertion of Lemma (7.3) is that these two computations give the same answer, i.e. $\tilde{\Lambda} \circ (H_r \times H_\Lambda)$ applied to the product symbol on $\tilde{\Gamma} \times \tilde{\Lambda}$ is the same as $H_r \times H_\Lambda$ applied to this symbol followed by $\tilde{\Lambda}$. One small adjustment has to be made in this argument. The computations of $\sigma(k)$ and $\sigma(\mu)$ both involve generic phase functions but the computation of the symbol of (7.7) involves a clean phase function with excess e ; so the factor of $(2\pi i)$ occuring in $\sigma(K\mu)$ is not the same as the product of the factors occuring in $\sigma(k)$ and $\sigma(\mu)$ i.e. we get

$$\sigma(K\mu) = (2\pi i)^{-\frac{e}{2}} \sigma(K) \sigma(\mu).$$

Finally a word regarding the Maslov factors. The Maslov factor involved in (7.7) is $\text{sgn}(\varphi + \psi)''_{\omega\omega}$, and the Maslov factors involved in $\sigma(k)$ and $\sigma(\mu)$ are $\text{sgn} \varphi''_{\theta\theta}$ and $\text{sgn} \psi''_{\xi\xi}$. Therefore the Maslov factor for $\sigma(K\mu)$ is the appropriate composite Maslov factor in view of (5.8).

Appendix

Suppose the periodic bicharacteristics are isolated, occur with distinct periods, and satisfy the genericity condition: $I - P_\gamma$ invertible, P_γ being the Poincaré map of the bicharacteristic γ . If $k\gamma$ is the k fold iterate of γ this means that $I - P_{k\gamma} = I - P_\gamma^k$ is invertible for all k ; so by (4.9) we can determine the numbers

$$|\det(I - P_\gamma^k)|, \quad k = 0, 1, 2, \dots, \quad (*)$$

for all k , from the spectrum of Q . To what extent do the numbers $(*)$ determine the eigenvalues of P_γ ? Being unable to answer this question satisfactorily ourselves we asked Harold Stark about it and obtained from him the following answer which he has kindly allowed us to reproduce in this appendix:

Theorem. *Let P be a $k \times k$ symplectic matrix having no roots of unity for eigenvalues. Then, knowing the numbers $\det(I - P^n)$ $n = 1, 2, 3, \dots$, one can determine*

- a) *the eigenvalues of P of absolute value $\neq 1$*
- b) *the eigenvalues of P^N for some integer N .*

In other words, one can determine the eigenvalues of P of absolute value $= 1$ up to multiplication by N -th roots of unity. Unfortunately, as will be seen below, there is no simple way of bounding N in b).

Corollary. *One can determine a) and b) just from knowing $|\det(I - P^n)|$ for $n = 1, 2, 3, \dots$*

Proof. Apply the theorem to $\begin{pmatrix} P & 0 \\ 0 & P \end{pmatrix}$.

§1. The proof of part a) of the theorem is rather easy. Namely, form the “generating function”:

$$f(t) = \sum_{n=0}^{\infty} \frac{\mu_{2n}}{\mu_n} t^n, \quad \mu_n = \det(I - P^n).$$

If $a_1, \dots, a_k$ are the eigenvalues of P , then

$$\begin{aligned} f(t) &= \sum_{n=0}^{\infty} \prod_{i=1}^k (1 + a_i^n) t^n = \frac{1}{1-t} + \sum_i \frac{1}{1-a_i t} + \sum_{i < j} \frac{1}{1-a_i a_j t} + \dots \\ &= \sum \frac{m(\omega)}{1-\omega t}, \end{aligned} \quad (1.1)$$

ω being one of the numbers on the list:

$$\prod_{i=1}^k a_i^{\varepsilon_i} \quad \varepsilon_i = 0, 1, \quad (1.2)$$

and $m(\omega)$ the number of times it occurs as such a product. We already see from (1.1) that there can only be a finite number of eigenvalue combinations having the given μ_n 's.

Let ω_0 be a number on the list (1.2) of minimum absolute value (ω_0 not necessarily unique). Suppose $\omega_0 = \prod a_i^{\delta_i}$, $\delta_i = 0, 1$. Consider the quotients ω/ω_0 . These are of the form

$$\prod a_i^{\varepsilon_i - \delta_i} = \prod_{\delta_i=0} a_i^{\varepsilon_i} \prod_{\delta_i=1} (a_i^{-1})^{1-\varepsilon_i}, \quad \varepsilon_i = 0, 1. \quad (1.3)$$

Let $b_i = a_i$ if $\delta_i = 0$, and $b_i = a_i^{-1}$ if $\delta_i = 1$. Then the numbers (1.3) are the set of all numbers

$$\prod b_i^{\varepsilon_i}, \quad \varepsilon_i = 0, 1. \quad (1.4)$$

We claim that knowing the products (1.4) one can determine the b_i 's of absolute value $\neq 1$. In fact the products (1.4) are all of absolute value ≥ 1 . Suppose

$$|b_1| = \dots = |b_s| = 1 \quad \text{and} \quad |b_{s+1}|, \dots, |b_k| > 1.$$

Consider among the products (1.4) the ones of absolute value $= 1$. These are the products

$$\prod_{i=1}^s b_i^{\varepsilon_i} \quad \varepsilon_i = 0, 1 \quad (1.5)$$

therefore we can extract these products from the list (1.4); and, in analogy with (1.1), determine the generating function

$$\sum_{i=1}^s \prod_{i=1}^s (1 + b_i^n) t^n.$$

We also know the generating function associated with the products (1.4) and comparing the coefficients of these two series, we get

$$\sum_{i=s+1}^k \prod_{i=s+1}^k (1 + b_i^n) t^n$$

so, again in analogy with (1.1), we know all the products

$$\prod_{i=s+1}^k b_i^{\varepsilon_i} \quad \varepsilon_i = 0, 1.$$

Among these products the ones of minimum absolute value must be b_i 's. Since each b_i is an a_i (P is a symplectic matrix so the inverse of an eigenvalue is an eigenvalue), we've located one eigenvalue, say a , of absolute value > 1 . Divide the n -th coefficient of (1.1) by $(1+a^n)(1+a^{-n})$ and repeat the argument with the remaining a_i 's. After a finite set of steps all the eigenvalues of absolute value $\neq 1$ can be determined.

§ 2. Given real numbers a and b let's say $|a-b| < \varepsilon \pmod{1}$ if there exists an integer n such that $|a-b-n| < \varepsilon$. Recall Kronecker's theorem:

Theorem. *Let $s_1, \dots, s_l$ be real numbers which are linearly independent over the rationals. Then, given any set of l real numbers $y_1, y_2, \dots, y_l$ and any $\varepsilon > 0$ there exists an integer N such that $|Ns_i - y_i| < \varepsilon \pmod{1}$ for $i = 1, \dots, l$.*

(See Hardy-Wright, [15].)

Now suppose there exist two $k \times k$ matrices P and Q such that for all n

$$\det(I - P^n) = \det(I - Q^n). \quad (2.1)$$

Let $a_1, \dots, a_k$ be the eigenvalues of P and $b_1, \dots, b_k$ the eigenvalues of Q . Assume as in §1 that there are no roots of unity among the a_i 's and b_i 's. Assume also (an assumption justified by §1) that the a_i 's and b_i 's are all of absolute value 1. Let b be a b_i . Then by (2.1),

$$\left| \frac{\prod (1 - a_i^n)}{1 - b^n} \right| \leq 2^{k-1} \quad \text{for all } n. \quad (2.2)$$

Our main result will be:

Lemma. *If (2.2) holds then some a_i is a power of b times a root of unity.*

Before proving this lemma (the proof is a little complicated) let us explain why Kronecker's theorem is useful for analyzing estimates of the type (2.2). Suppose that not only is no a_i a power of b times a root of unity, but that the a_i 's and b together form a multiplicatively independent set, i.e. there are no non-trivial products of the form

$$a_1^{l_1} a_2^{l_2} \dots a_k^{l_k} b^m = 1.$$

Then, by Kronecker's theorem, for every $\varepsilon > 0$, we can find an N such that

$$|a_i^N + 1| < \varepsilon \quad \text{and} \quad |b^N - 1| < \varepsilon.$$

Comparing with 2.2 we get a contradiction

$$\left| \frac{\prod (1 - a_i^N)}{1 - b^N} \right| > \frac{(2 - \varepsilon)^k}{\varepsilon}$$

for ε small; therefore the a_i 's and b must be multiplicatively dependent.

The proof of the lemma will be a refinement of this argument. Consider the numbers

$$1, \theta_i = \frac{1}{2\pi\sqrt{-1}} \log a_i, \quad \theta = \frac{1}{2\pi\sqrt{-1}} \log b. \quad (2.3)$$

The θ_i 's and θ are, of course, only determined modulo 1. Assume that $1, \theta, \theta_1, \dots, \theta_l$ are linearly independent over the rationals, and the other θ_i 's are linear combinations of them with rational coefficients. Then, for all $i = 1, \dots, k$ we can write

$$\theta_i = \frac{r_i}{q} + \frac{p_i}{q} \theta + \sum_{j=1}^l \frac{p_{ij}}{q} \theta_j. \quad (2.4)$$

Multiplying (2.4) by $n = Nq$ we get

$$n\theta_i \equiv p_i N \theta + \sum_{j=1}^l p_{ij} N \theta_j \pmod{1}. \quad (2.5)$$

Let μ be a fixed integer (which in the argument below will be either 0 or 1). Choose real numbers $x_1, \dots, x_l$ such that for each $i = 1, \dots, k$ for which the p_{ij} 's are not all zero

$$\sum_{j=1}^l p_{ij} x_j + \frac{p_i \mu}{q} \not\equiv 0 \pmod{1}. \quad (2.6)$$

(Such a choice of x_i 's is clearly possible.) By Kronecker we can choose $N = N(\varepsilon)$ so that

$$\left| N\theta - \frac{\mu}{q} \right| < \varepsilon \pmod{1} \quad (2.7)$$

and

$$|N\theta_j - x_j| < \varepsilon \pmod{1} \quad j = 1, \dots, l. \quad (2.8)$$

We will now show that if the lemma is not true (2.2) gets contradicted for this choice of N and n . First let $\mu = 0$ in (2.6) and (2.8). Then

$$n\theta \equiv O(\varepsilon) \pmod{1}$$

by (2.7) and

$$n\theta_i \equiv \sum p_{ij} x_j + O(\varepsilon) \pmod{1}$$

by (2.7) and (2.8). Suppose that for all i , some $p_{ij} \neq 0$. Then for all i , $1 - a_i^n = 1 - e^{2\pi\sqrt{-1}n\theta_i}$ is bounded away from 0 by (2.6); however $1 - b^n = 1 - e^{2\pi\sqrt{-1}n\theta}$ is of order $O(\varepsilon)$. This contradicts (2.2) for small ε and therefore implies that for some i all $p_{ij} = 0$. Note that for such an i $q\theta_i = p_i\theta \pmod{1}$ by (2.4) so $a_i^q = b^{p_i}$, which is nearly the result we want. In fact the proof of our lemma would be finished if we could show q divides p_i . To show this, let $\mu = 1$ in (2.7). Then for every i for which all $p_{ij} = 0$,

$$n\theta_i \equiv \frac{p_i}{q} + O(\varepsilon) \pmod{1} \quad \text{by (2.5),}$$

and for the other θ_i 's

$$n\theta_i \equiv \sum_{j=1}^l p_{ij} x_j + \frac{p_i \mu}{q} + O(\varepsilon) \pmod{1}.$$

We still have

$$n\theta = Nq\theta \equiv O(\varepsilon) \pmod{1}$$

by (2.7). It is clear that for some i , for which all the p_{ij} 's vanish p_i/q must be an integer; otherwise as before all the terms in the numerator of (2.2) will be bounded away from 0 and the denominator will be of order ε . This concludes the proof of the lemma.

To conclude the proof of the theorem, go back to (2.1) and consider the eigenvalues of P^r and Q^r i.e.

$$a_1^r, \dots, a_k^r$$

and

$$b_1^r, \dots, b_k^r.$$

For sufficiently large r every number in the second set, raised to some power, equals a number in the first set and visa versa. It is clear this can only happen if the two sets are identical.

References

1. Andersson, G. K.: Analytic wave front sets for solutions of linear differential equations of principal type. *Trans. Am. Math. Soc.* **177**, 1–27 (1973)
2. Arnol'd, V. I.: On a characteristic class entering in quantization conditions. *Funct. Anal. Appl.* **1**, 1–13 (1967)
3. Atiyah, M. F., Bott, R.: A Lefschetz fixed point formula for elliptic complexes I. *Ann. of Math.* **86**, 374–407 (1967)
4. Atiyah, M. F., Bott, R., Patodi, V. K.: On the heat equation and the index theorem. *Inventiones math.* **19**, 279–330 (1973)
5. Bott, R.: On the iteration of closed geodesics and the Sturm intersection theory, *Comm. Pure Appl. Math.* **9**, 176–206 (1956)
6. Chazarain, J.: Formule de Poisson pour les variétés riemanniennes. *Inventiones math.* **24**, 65–82 (1974)
7. Colin de Verdière, Y.: Spectre du laplacien et longueurs des géodésiques périodiques II. *Comp. Math.* **27**, 159–184 (1973)
8. Cotsafis, M.: Une propriété des orbites périodiques des systèmes hamiltoniens non-linéaires. *C. R. Acad. Sc. Paris* 275, Série A 911–914 (1973)
9. Duistermaat, J. J., Hörmander, L.: Fourier integral operators II. *Acta Math.* **128**, 184–269 (1972)
10. Duistermaat, J. J.: Fourier Integral Operators. Courant Institute Lecture Notes, New York 1973
11. Duistermaat, J. J., Guillemin, V. W.: The spectrum of positive elliptic operators and periodic geodesics. *Proc. A.M.S. Summer Institute on Differential Geometry, Stanford 1973* (to appear)
12. Duistermaat, J. J.: On the Morse index in variational calculus. To appear in *Advances in Math.*
13. Gelfand, I. M., Shilov, G. E.: *Generalized Functions, I*. New York: Academic Press 1964
14. Guillemin, V., Sternberg, S.: *Geometric Asymptotics*. A.M.S. Publications (in press)
15. Hardy, G. E., Wright, E. M.: *An Introduction to the Theory of Numbers*, 4th ed. Oxford: Clarendon Press 1960
16. Hörmander, L.: The spectral function of an elliptic operator. *Acta Math.* **121**, 193–218 (1968)
17. Hörmander, L.: Fourier integral operators I. *Acta Math.* **127**, 79–183 (1971)
18. Minakshisundaram, S., Pleijel, Å.: Some properties of the eigenfunctions of the Laplace operator on Riemannian manifolds. *Canadian J. Math.* **1**, 242–256 (1949)
19. Nirenberg, L.: Lectures on Linear Partial Differential Equations, *Regional Conference Series in Mathematics*, No 17. Conf. Board of the Math. Sc. of the A.M.S., 1972
20. Sato, M.: Regularity of hyperfunction solutions of partial differential equations. *Proc. Nice Congress*, Vol. 2, pp. 785–794. Paris: Gauthiers-Villars 1970
21. Sato, M., Kawai, T., Kashiwara, M.: *Microfunctions and Pseudo-Differential Equations*. *Lecture Notes in Mathematics* **287**, pp. 265–529. Berlin-Heidelberg-New York: Springer 1973

22. Seeley, R. T.: Complex powers of an elliptic operator. A.M.S. Proc. Symp. Pure Math. **10**, 288–307 (1967). Corrections in: The resolvent of an elliptic boundary problem. Am. J. Math. **91**, 917–919 (1969)
23. Serre, J.-P.: A Course in arithmetic. Berlin-Heidelberg-New York: Springer 1973
24. Weinstein, A.: Fourier integral operators, quantization and the spectra of Riemannian manifolds. To appear in the Proc. of the C.N.R.S. Colloque de Géométrie Symplectique et Physique Mathématique. Aix-en-Provence, June 1974
25. Klingenberg, W., Takens, F.: Generic properties of geodesic flows. Math. Ann. **197**, 323–334 (1972)
26. Hörmander, L.: Linear differential operators. Proc. Nice Congress, Vol. I, pp. 121–133. Paris: Gauthiers-Villars 1970

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Note Added in Proof. Colm de Verdière (Séminaire Goulaouic-Lions-Schwartz 1974–75) has shown that for $P = (-\Delta + c)^{\frac{1}{2}}$ the number σ_m (6.16) can be identified with the Morse index for periodic geodesics by writing $U(t, x, y)$ as an approximate Feynman integral.

